

REVIEW

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Recent advances in ferroelectric materials, devices, and in-memory computing applications

Hwiho Hwang^{1†} , Sangwook Youn^{1†} and Hyungjin Kim^{1*}

Abstract

Ferroelectric memories have undergone a transformative evolution from conventional perovskite-based materials to modern fluorite-structured ferroelectrics, driven by the pursuit of scalable, low-power, and CMOS-compatible non-volatile memory solutions. The observation of ferroelectricity in nanoscale HfO₂-based films has enabled integration with CMOS-compatible processes, providing advantages such as potential scalability, low power consumption, and non-volatility, while facilitating continued scaling and high-density integration. Leveraging established materials infrastructure in the semiconductor industry, hafnia-based ferroelectrics have been incorporated in various memory architectures, including ferroelectric random-access memory (FeRAM), ferroelectric tunnel junctions (FTJs), ferroelectric field-effect transistors (FeFETs), and ferroelectric memcapacitors (FeCAPs). Beyond conventional non-volatile storage, these devices have also emerged as promising building blocks for in-memory computing applications, including neuromorphic systems, hardware security primitives, and associative memory. In this review, we explore the historical development of ferroelectric memories from a materials–device co-design perspective, examine recent advances in device architectures and in-memory computing applications, and discuss the remaining challenges in endurance, retention, variability, and scaling. Finally, we propose future research directions that integrating material innovation, interface engineering, and circuit-level optimization to realize the full potential of ferroelectric memories in next-generation computing platforms.

Keywords Ferroelectric thin films, Non-volatile memory devices, In-memory computing, Neuromorphic computing, Hardware security

1 Introduction

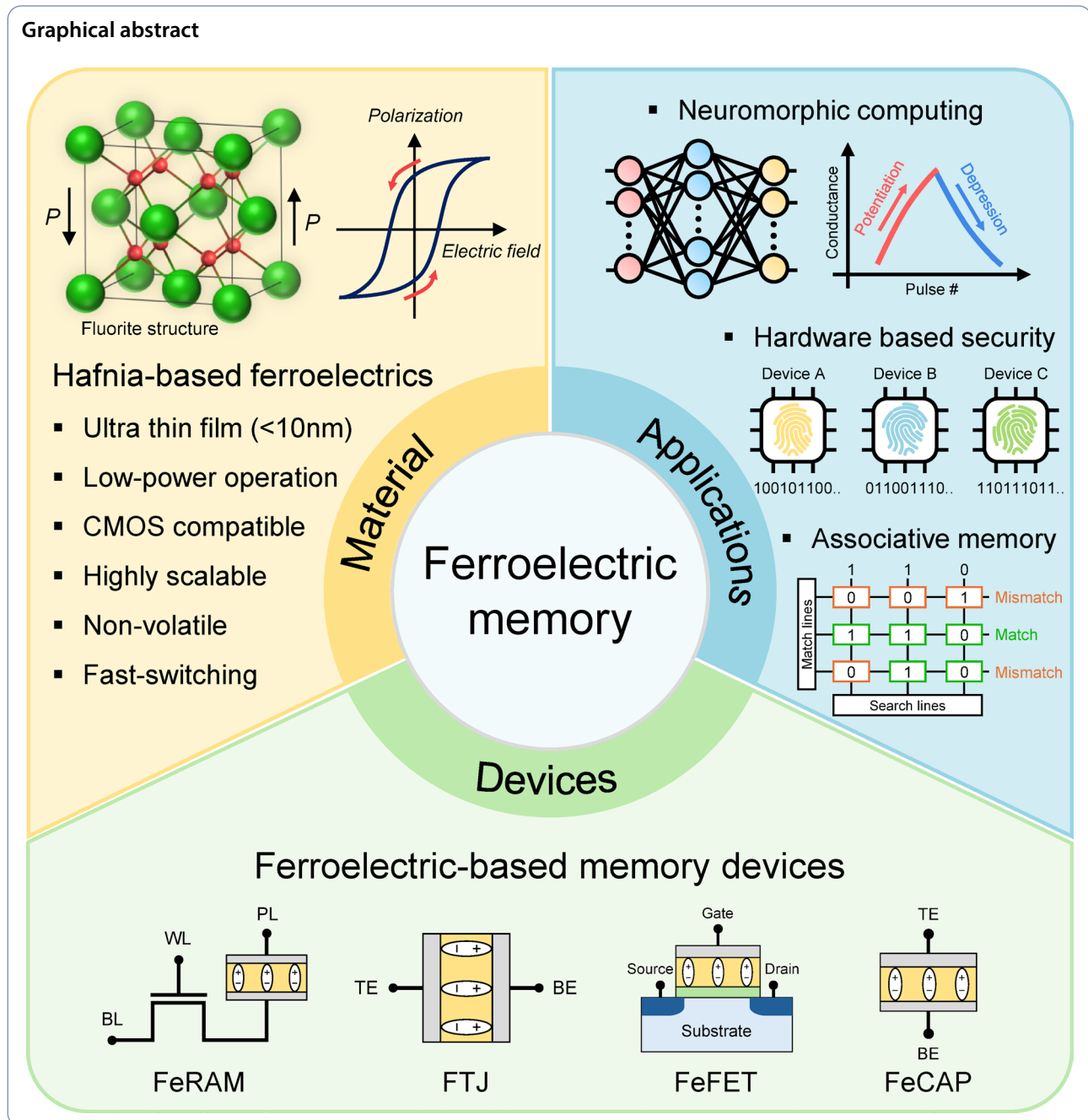
Ferroelectricity is the property of having multiple stable polarization states, arising from a non-centrosymmetric crystal structure. This feature enables spontaneous polarization without an external electric field, and allows reversible switching under an applied field. Owing to these properties, ferroelectric (FE) materials have long been regarded as promising candidates for next-generation nonvolatile memory devices [1–3]. However, the advantageous characteristics of conventional perovskite-based ferroelectrics could not be preserved upon scaling to nanoscale films integrated on silicon, due to

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Graphical abstract



polarization degradation, poor CMOS compatibility, and reliability concerns such as fatigue and imprint. These limitations have long hindered the realization of practical ferroelectric devices in mainstream semiconductor technology. A breakthrough occurred in 2011 with the discovery of ferroelectricity in 10 nm-thick Si-doped HfO_2 thin films, where top electrode (TE) capping followed by annealing and cooling induced interfacial stress and confinement effects that stabilized the metastable orthorhombic phase ($\text{Pca}2_1$) responsible for ferroelectricity. Importantly, HfO_2 had already been widely adopted

in CMOS technology through atomic layer deposition (ALD), and this finding consequently sparked intense interest in ferroelectric memories owing to their CMOS compatibility and scalability toward advanced technology nodes [4–10]. Representative device types include the one-transistor-one-capacitor (1T-1C) ferroelectric random-access memory (FeRAM) [11], ferroelectric tunnel junction (FTJ) [12], ferroelectric field-effect transistor (FeFET) [13, 14], and ferroelectric memcapacitor (FeCAP) [15].

Fluorite-structured ferroelectrics have emerged as a promising material platform for ferroelectric memory devices. These materials are compatible with both silicon and a wide range of metal electrodes used in the semiconductor industry. Notably, HfO_2 has long been employed as a high- k gate dielectric in CMOS transistors, while ZrO_2 , one of the most promising dopants for stabilizing ferroelectricity in HfO_2 , has been widely used in Dynamic random-access memory (DRAM) cell capacitors. Thanks to this strong compatibility with CMOS process technologies, functional devices including FeRAM and FeFET have already been demonstrated, while research efforts on FTJ and FeCAP devices are rapidly accelerating. This unique position at the intersection of well-established materials infrastructure and novel ferroelectric functionality has driven intense development in the field. Despite their promise, HfO_2 based ferroelectrics face persistent challenges in reliability metrics such as endurance and retention, as well as in suppressing competing non-ferroelectric phases. Among these, the quality of the interface between the ferroelectric film and its surrounding layers is one of the most critical issues, which can strongly influence switching behavior and device stability. Materials and process optimization remain necessary to address these issues. Another major limitation is the restricted stabilization of the ferroelectric orthorhombic phase at very thin film thicknesses, which is essential for scaling to future device generations. Addressing these challenges requires an integrated approach encompassing materials design, interface engineering, and process refinement. Beyond their role in conventional non-volatile memory (NVM), HfO_2 -based ferroelectrics have been utilized for a variety of in-memory computing paradigms aimed at overcoming the limitations of traditional von Neumann architectures. Their intrinsic domain-switching behavior allows partial polarization states to be accessed, enabling multi-level conductance for neuromorphic computing. The stochastic distribution of initial ferroelectric domains can serve as a high-entropy physical source for hardware security primitives such as physically unclonable functions (PUFs). Furthermore, their non-volatility and fast switching make them suitable for associative memory architecture. These properties make HfO_2 -based ferroelectrics a key enabler for emerging computing applications, with growing research interest in both material and device-level optimization.

In this paper, we review the historical development of ferroelectric memories from a material–device integration perspective, with a particular focus on hafnia-based ferroelectrics. Section 2 traces the evolution of ferroelectric materials, from the discovery of ferroelectricity and early perovskite ferroelectrics to the recent emergence of hafnia-based fluorite ferroelectrics. Section 3 discusses HfO_2 -based ferroelectric memory architectures and their

technological progress. Section 4 reviews applications of HfO_2 -based ferroelectric devices in in-memory computing, including neuromorphic computing, hardware security, and associative memory. Section 5 addresses the current challenges of ferroelectric memories and explores potential solutions. Finally, we provide a forward-looking perspective on future research strategies that integrate material innovations with device and architectural design for next-generation memory technologies.

2 Ferroelectric materials

2.1 Historical development of ferroelectric materials

The concept of ferroelectricity, characterized by two stable spontaneous polarization states, was first introduced by Erwin Schrödinger in 1912 [16]. It was experimentally demonstrated for the first time in 1921 by Valasek using Rochelle salt [17–19]. During the following decade, research on Rochelle salt continued, but its ferroelectric properties were found to be highly sensitive to environmental factors such as humidity. In 1935, Busch et al. discovered ferroelectricity in KH_2PO_4 , which exhibited greater chemical stability and therefore stimulated further investigation [20]. However, its Curie temperature (T_c) was lower than room temperature, which limited its practical use. In the early 1940s, during World War II, the perovskite-structured BaTiO_3 was discovered. This material offered both room-temperature stability and a high dielectric constant ($\epsilon_r > 1000$), marking a breakthrough that led to intensive research on perovskite-based ceramics and greatly advanced the understanding and application of ferroelectric phenomena [21–23]. Structurally, perovskite-structured ferroelectrics possess the general chemical formula ABO_3 , where two cations of different sizes occupy the A-site (cube corners) and B-site (cube center), respectively. The B cation resides at the center of an octahedron formed by six oxygen atoms. Ferroelectricity arises when the B cation is slightly displaced from the octahedral center, generating an electric dipole, and when these dipoles align in a common direction, spontaneous polarization emerges. Following the discovery of BaTiO_3 , the Landau–Ginzburg–Devonshire (LGD) phenomenological model was developed to describe ferroelectricity and its phase transitions [24–27]. In the subsequent decades, extensive research was devoted to perovskite-structured ferroelectrics, with one of the most significant findings being the $\text{Pb}(\text{Zr,Ti})\text{O}_3$ (PZT) solid solution. PZT exhibits a high T_c of approximately 400 °C and a large remanent polarization (P_r) of 10–40 $\mu\text{C cm}^{-2}$. Furthermore, by tuning the Zr/Ti ratio to form a morphotropic phase boundary, its piezoelectric properties are significantly enhanced, leading to major advancements in ferroelectric research [28–30]. In 1960, Cochran and Anderson independently proposed the concept of the soft mode, in which a specific phonon mode at the center

History of ferroelectric memories

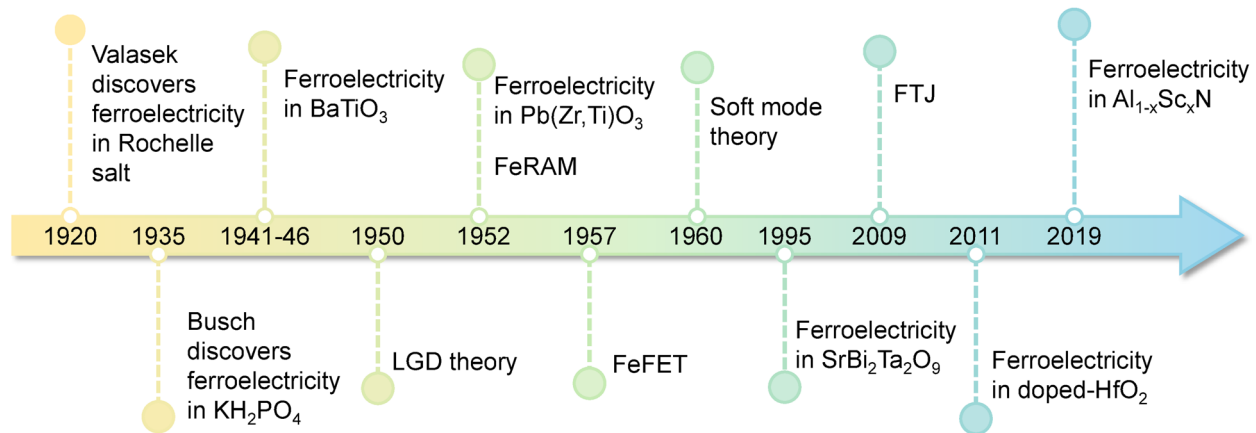


Fig. 1 Historical development of ferroelectric (FE) materials and devices

of the Brillouin zone gradually softens with decreasing temperature, causing its frequency to approach zero. As a result, atoms remain in a displaced position rather than returning to their equilibrium sites, leading to the emergence of spontaneous polarization. This phenomenon established the fundamental basis for understanding displacive phase transitions in ferroelectrics [31–33].

Also, FeRAM and FeFET were first proposed in the 1950s [34] and 1960s [35], respectively. These devices demonstrated the potential to exploit the fast polarization switching speed and low power consumption of ferroelectric materials for memory applications, generating significant interest in their integration into memory and semiconductor devices. However, conventional perovskite ferroelectrics such as PZT and BaTiO₃ exhibited severe fatigue issues in FeRAM. In 1995, the layered perovskite SrBi₂Ta₂O₉ (SBT) was reported for the first time [36], offering high fatigue resistance and overcoming the lifetime limitations of PZT-based FeRAM, which accelerated its commercialization. In 2009, Garcia et al. reported the first FTJ based on BaTiO₃ [37], introducing a new ferroelectric memory concept that employed a few-nanometer-thick ferroelectric barrier instead of the thick films used in FeRAM, enabling low-power, high-speed, non-volatile operation. In 2011, ferroelectricity was first observed in silicon-doped HfO₂ thin films with a fluorite structure [38], marking a paradigm shift in ferroelectric research. Given that HfO₂ was already widely used as a high- k dielectric in CMOS processes, its compatibility with existing semiconductor manufacturing and ability to retain ferroelectricity in ultra-thin films (<10 nm) made it highly attractive for device scaling. This discovery triggered a surge of research into its integration in logic and memory devices. In 2019, Fichtner et al. reported ferroelectricity in Sc-doped AlN (Sc_xAl_{1-x}N)

Table 1 Material properties of representative ferroelectric materials, including Pb(Zr,Ti)O₃, SrBi₂Ta₂O₉, doped HfO₂, and (Al, Sc)N

	Pb(Zr, Ti) O ₃	SrBi ₂ Ta ₂ O ₉	Doped HfO ₂	(Al, Sc)N
P_r [$\mu\text{C}/\text{cm}^2$]	10–40	5–10	10–40	80–110
E_c [kV/cm]	50–70	30–50	800–2000	2000–5000
ϵ_r	~400	~200	~30	~25
Min. thickness [nm]	50	–	< 5	< 10

[39], exhibiting a T_c above 1000 °C and a P_r of ~110 $\mu\text{C}/\text{cm}^2$, thereby introducing a high-temperature-stable ferroelectric material with potential to expand the application range of ferroelectric devices. Figure 1 summarizes the historical development of ferroelectric memories, while Table 1 presents the material properties of representative ferroelectrics, including the PZT, SBT, doped HfO₂, and (Al,Sc)N.

2.2 Hafnia-based ferroelectricity

As discussed earlier, Börscke et al. first reported ferroelectricity in Si-doped HfO₂ thin films, which initiated a new research direction in hafnia-based ferroelectrics [38]. This breakthrough attracted significant interest, as HfO₂ was already widely used as a high- k gate dielectric in CMOS technology thanks to the ALD deposition. Its compatibility with conventional semiconductor fabrication processes, ability to maintain ferroelectricity even in ultrathin (< 10 nm) films, and advantages for device scaling established hafnia-based ferroelectrics as promising candidates for next-generation nonvolatile memory and logic applications. In bulk at room temperature, HfO₂ is monoclinic and non-ferroelectric, transforming to tetragonal and cubic phases at elevated temperatures. During deposition and thermal treatment, a high-temperature

tetragonal phase may form and, upon cooling, particularly under a TE capping effect, transform into the non-centrosymmetric orthorhombic phase with switchable polarization. Figure 2a illustrates the tetragonal to orthorhombic phase transformation in Si:HfO₂ thin films, yielding two polarization states (red: oxygen; white: hafnium). Figure 2b shows polarization–electric field (P – E) curves and AC dielectric constant–electric field characteristics for different Si contents, indicating that ferroelectric properties vary with dopant content [38]. In particular, as the silicon content increases, the polarization gradually shifts from FE to antiferroelectric (AFE), which is also evidenced by the dielectric constant exhibiting a single peak in the FE state and two peaks in the

AFE state. Here, P_r , defined as the magnitude of the spontaneous polarization that remains after the external electric field is removed from the ferroelectric layer, reflects the strength of the ferroelectric behavior. A large P_r not only signifies strong ferroelectricity but also enhances the charge-sensing signal, read current margin, and on/off ratio in FeRAM, FeFET, and FTJ devices. Consequently, research on hafnia-based ferroelectrics has focused not only on understanding their ferroelectric characteristics but also on achieving higher P_r values.

Hafnia-based ferroelectrics with various dopants (e.g., Zr, Gd, Y, La) have been reported to stabilize the orthorhombic phase [40–45]. Figure 2c compares P – E curves of TiN/8–10 nm X:HfO₂ or HfZrO₄/TiN capacitors with

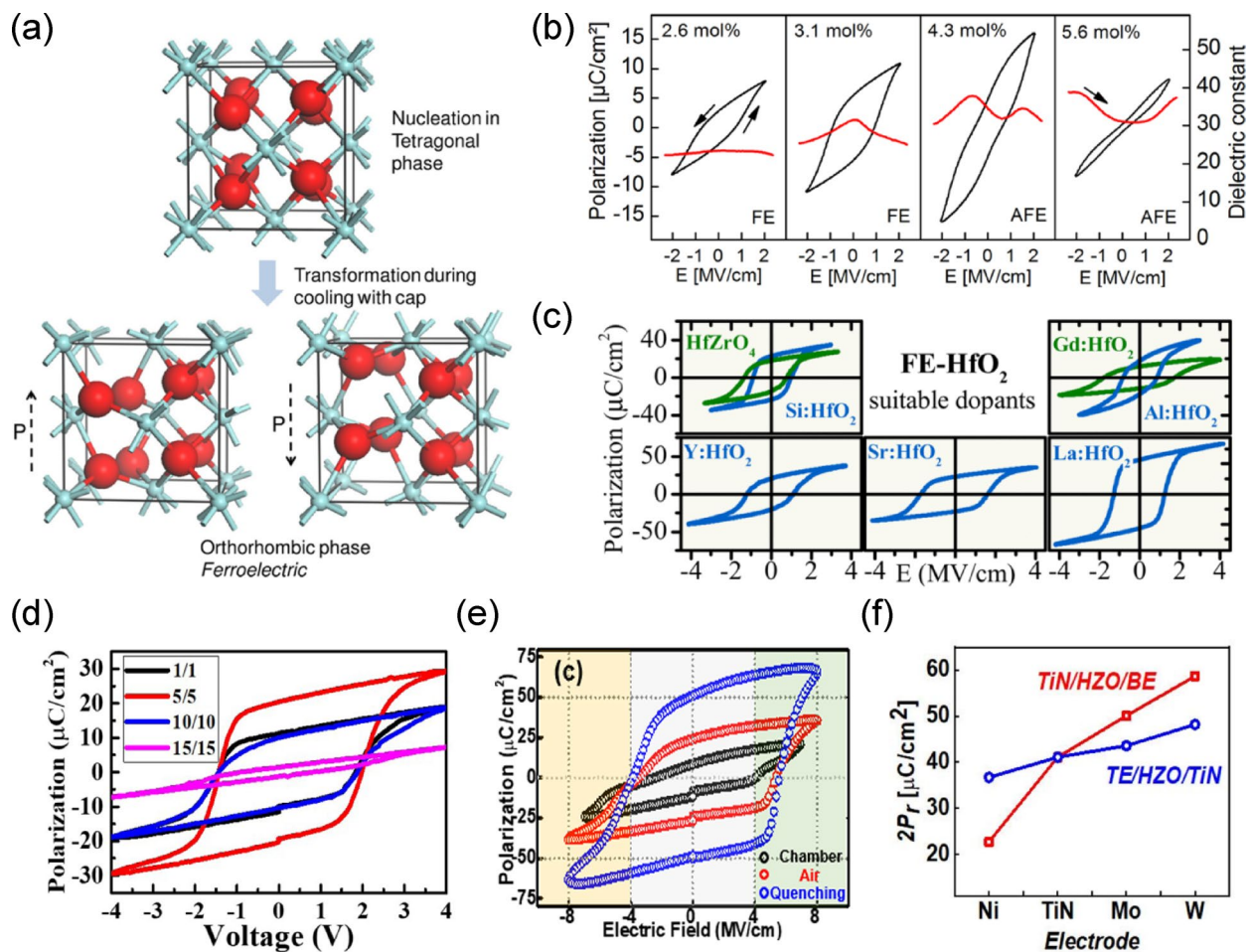


Fig. 2 Hafnia-based ferroelectrics. **a** Schematic illustration of the phase transformation from the tetragonal to the orthorhombic phase in Si:HfO₂ thin films, leading to the formation of a ferroelectric phase with two distinct polarization states (red atoms: oxygen, white atoms: hafnium). **b** P – E curves (black, left y-axis) and AC dielectric constant–electric field characteristics (red, right y-axis) for different SiO₂ admixture amounts. As the silicon content increases, the polarization curves transition from FE to antiferroelectric (AFE), with the dielectric constant shifts from exhibiting a single peak to showing two distinct peaks.

Reproduced with permission from Ref [38]. Copyright 2011, American Institute of Physics. **c** P – E curves of TiN/8–10 nm X:HfO₂ or HfZrO₄/TiN capacitors with various dopants X. Reproduced with permission from Ref [42]. Copyright 2013, IEEE. **d** P – V curves of TiN/Zr-doped HfO₂ (HZO)/TiN capacitors for different ALD cycle ratios of HZO deposition. Reproduced with permission from Ref [53]. Copyright 2019 IEEE. **e** P – E characteristics of Al:HfO₂ ferroelectric capacitors for different cooling rates. Reproduced with permission from Ref [56]. Copyright 2020 IEEE. **f** Comparison of remanent polarization for HZO ferroelectric capacitors with different metal electrodes. Reproduced with permission from Ref [59]. Copyright 2021, IEEE

different dopants X and notably, the La-doped HfO_2 thin film exhibited the P_r reaching approximately $45 \mu\text{C}/\text{cm}^2$ (Müller et al. [42]). It has also been reported that adjusting the dopant concentration influences the ferroelectric properties [46]. Furthermore, hafnia-based ferroelectric films have been shown to retain ferroelectricity at thicknesses down few nanometers [47]. In Zr-doped HfO_2 , particularly $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ (HZO) films, a counterintuitive enhancement of polarization has been observed when the thickness was reduced to ultra-thin regime of 1 nm [48]. This finding shows that the enhancement effect persists down to the thickness of at least two fluorite-structure unit cells, overcoming the deleterious depolarization field that typically arises in the ultrathin regime of conventional perovskite films [49]. By mitigating this scaling limitation, it demonstrates the potential for realizing next-generation, ultra-scaled ferroelectric transistors. While many doped HfO_2 films require high-temperature annealing, which limits the integration of ferroelectric circuits into back-end-of-line (BEOL) processes, HZO has shown ferroelectricity within a low thermal budget (400–450 °C), including cases where crystallization occurs during TE TiN ALD without an additional heat treatment [50, 51]. Kim et al. further showed that optimizing the TiN TE thickness in TiN/HZO/TiN capacitors enabled high $2P_r \sim 45 \mu\text{C}/\text{cm}^2$ at 400 °C [52].

Also, several studies have been conducted to achieve higher P_r values by optimizing the ferroelectric layer through deposition methods, annealing conditions, and metal electrode engineering. Liao et al. engineered the grain size of HZO thin films by optimizing the HfO_2 -to- ZrO_2 cycle ratio during ALD, achieving a $2P_r$ value of $41 \mu\text{C}/\text{cm}^2$. The corresponding polarization–voltage (P – V) curves of TiN/HZO/TiN capacitors for different ALD cycle ratios are shown in Fig. 2d [53]. Peng et al. demonstrated that employing a superlattice HZO thin film in metal–ferroelectric–metal (MFM) capacitors increased P_r compared to conventional HZO films, while simultaneously reducing leakage current and improving endurance, achieving over $>10^{12}$ switching cycles [54]. Zhou et al. investigated the effect of annealing temperature on Al-doped HfO_2 (HAO) thin films, varying the temperature from 600 to 800 °C. By optimizing the annealing temperature to 800 °C, they engineered the crystallization of the HAO film and achieved a high P_r value of $23.7 \mu\text{C}/\text{cm}^2$ [55]. Ku et al. analyzed the influence of post-cooling processes after annealing the HAO ferroelectric layer, comparing chamber cooling, air cooling, and rapid quenching in deionized water. The corresponding P – E curves for each cooling condition are shown in Fig. 2e [56]. They found that rapid quenching induced greater stress/strain in the HAO films, which promoted the stabilization of the orthorhombic phase. As a result, a drastic increase in

P_r was observed, with $2P_r$ values approaching $\sim 100 \mu\text{C}/\text{cm}^2$.

Hoffmann et al. compared Gd-doped HfO_2 ferroelectric thin films using TiN and TaN electrodes. When TaN was used, a higher P_r of $35 \mu\text{C}/\text{cm}^2$ was obtained compared to TiN. This enhancement was attributed to the fact that, during thermal treatment, the TaN electrode underwent more extensive interfacial oxidation than TiN, which increased the concentration of oxygen vacancies in the HfO_2 layer. The increased vacancy concentration lowered the total energy of the orthorhombic phase in the lattice, thereby stabilizing the ferroelectric phase and enabling a higher P_r [41]. Goh et al. investigated the ferroelectric properties of a 4.5 nm HZO film with a TiN TE using W, Mo, Pt, TiN, and Ni as bottom electrodes (BEs). Among these, the W yielded the highest value, with a $2P_r$ of $62.4 \mu\text{C}/\text{cm}^2$ [57]. Similarly, Lee et al. examined the impact of various electrode materials for both the top and bottom contacts in HZO film. As shown in Fig. 2f, W electrodes produced the highest P_r values for both configurations compared to other electrodes. This enhancement was because of the lower coefficient of thermal expansion (CTE) of tungsten, which induced greater tensile stress in the HZO film during annealing, thereby increasing the orthorhombic phase fraction and improving ferroelectric properties [58].

3 Ferroelectric-based memory devices

3.1 Ferroelectric random-access memory (FeRAM)

FeRAM is a non-volatile memory based on a 1T-1C architecture, in which the dielectric layer of the capacitor is replaced with a ferroelectric material. Figure 3a shows the schematic of a FeRAM cell consisting of an access transistor and a FeCAP. During a write operation, the word line (WL) is activated to turn on the transistor, and a voltage is applied between the plate line (PL) and bit line (BL) to set the polarization direction of the ferroelectric layer. For readout, a voltage pulse is applied to the PL, inducing a polarization reversal charge at the BL according to the stored polarization state. The resulting signal, which differs depending on the polarization direction, is sensed by the sense amplifier (SA) to distinguish between logic “0” and “1”. Since polarization switching occurs during reading, FeRAM exhibits a destructive-read characteristic, requiring a write-back process after each read operation. The concept of FeRAM was first proposed in the 1950s [34], and with the maturation of planar ferroelectric PZT thin-film processing, commercial products emerged in the early 1990s for smart cards, consumer electronics, and industrial control systems [59]. However, perovskite-based FeRAM required relatively large cell structures to ensure sufficient capacitance, and its thin-film fabrication process posed integration challenges,

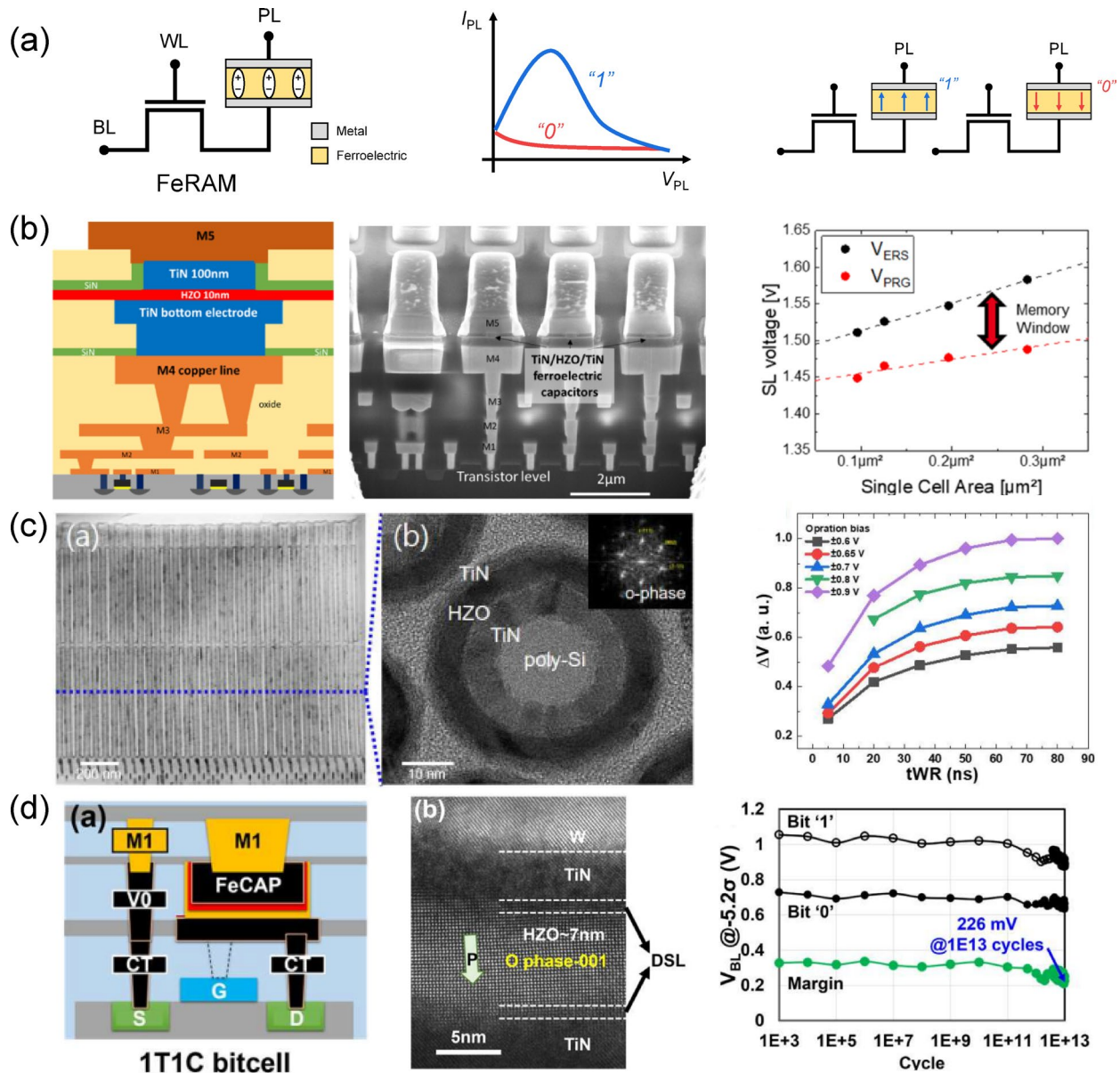


Fig. 3 FeRAM. **a** Schematics of 1T-1C FeRAM structure with I_{PL} - V_{PL} read characteristics and two polarization states. **b** Schematic of BEOL-compatible FeRAM cell co-integrated with 130 nm CMOS and fabricated with TiN/HZO/TiN ferroelectric capacitors (left panel), cross-sectional SEM image showing the capacitor stack integrated above the transistor level (middle panel), and measured source line voltage versus single-cell area demonstrating the memory window scaling (right panel).

Reproduced with permission from Ref [63]. Copyright 2019, IEEE. **c** Cross-sectional TEM image of a 1T-1C FeRAM with ultra-thin 5 nm HZO (left panel), high-resolution TEM and FFT image showing orthorhombic phase stabilization in TiN/HZO/TiN/poly-Si structure (middle panel), and measured ΔV as a function of write pulse width (t_{WR}) for various operation voltages (right panel). Reproduced with permission from Ref [66]. Copyright 2021, IEEE. **d** Schematic of a 1T-1C FeCAP bitcell for 3D integration (left panel), cross-sectional high-resolution TEM image showing a 7 nm HZO layer with orthorhombic phase orientation in the TiN/HZO/TiN stack (middle panel), and measured sensing voltage margin as a function of endurance cycles at 125 °C ($> 1E13$ cycles) (right panel). Specifically, two defect shielding layers (DSLs) are deposited on both sides of the HZO film to eliminate excess charged defects at the interfaces, thereby suppressing leakage and improving endurance. Reproduced with permission from Ref [70]. Copyright 2025, IEEE

confining its use to low-density, application-specific markets [60].

The introduction of CMOS-compatible HfO_2 -based ferroelectrics revitalized FeRAM scaling by enabling three-dimensional (3D) capacitor integration. In 2014, a deep-trench capacitor structure utilizing Al-doped HfO_2

demonstrated the potential for 3D-integrated non-volatile memory [61]. This progress advanced further in 2019, when Francois et al. demonstrated that a TiN/HZO/TiN ferroelectric capacitor could be monolithically integrated into the BEOL of a 130 nm CMOS process, marking a significant advance in embedded FeRAM scalability [62].

As shown in Fig. 3b (left panel), the cross-sectional schematic highlights the placement of ferroelectric capacitors between metal interconnect layers (M4 and M5), while the SEM image (middle panel) confirms successful fabrication of nanoscale capacitors down to 300 nm. Also, the electrical characterization (right panel) reveals stable memory windows as the cell area is scaled, with erase and program voltages maintaining clear separation even at sub- $1\mu\text{m}^2$ dimensions. Overall, these results demonstrate that BEOL-compatible HZO-based FeCAPs can achieve aggressive cell scaling while preserving robust ferroelectric properties, enabling high-density, low-power embedded non-volatile memory without additional process complexity.

Subsequently, several $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ -based 1 T-1C FeRAM arrays have been demonstrated, offering low-voltage operation and high endurance [63, 64]. Nevertheless, when scaled down to ultra-thin films, FeRAM characteristics are prone to degradation due to depolarization fields, interfacial dead layers, and stabilization of competing non-ferroelectric phases. In 2021, Sung et al. reported an ultra-thin HZO study that directly addressed these limitations [65]. As shown in the left panel of Fig. 3c, a cross-sectional image reveals the vertical storage node structure with a 1z nm MFM capacitor. The middle panel of Fig. 3c presents High-resolution transmission electron microscopy (HR-TEM) and Fast Fourier transform (FFT) patterns confirming the stabilization of the orthorhombic phase in a TiN/HZO (5 nm)/TiN/poly-Si stack. The right panel of Fig. 3c plots the dependence of ΔV on operating bias and write-pulse width, demonstrating that even at low voltages, a sufficient memory window can be maintained by shortening the pulse width. Overall, these results prove that reliable ferroelectric switching and fast write operations are achievable in 5 nm-thick HZO, supporting the feasibility of ultra-low-voltage, high-speed FeRAM scaling for future applications. In 2022, Toprasertpong et al. highlighted further reduced operating voltage, improved breakdown tolerance, and endurance exceeding 10^{12} cycles for HZO-based FeRAM [66], while Haratipour et al. showcased a 28 nm-class, ultra-high-density, high-speed FeRAM targeting next-generation embedded memory [67]. In parallel, a BEOL-integrated 16 kb $\text{HfO}_2\text{:Si}$ FeRAM array achieved 4 ns programming, 10^7 -cycle endurance, 125 °C retention, and solder-reflow compatibility [68].

Most recently, a 3D trench-type 1 T-1C FeRAM integrating a 7 nm-thick $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ layer with dual-side defect shielding layers (DSLs) has been demonstrated, achieving significant reliability improvements for embedded non-volatile memory applications [69]. As shown in the left panel of Fig. 3d, the cross-sectional schematic illustrates the vertical trench capacitor integrated above the access transistor, with DSLs placed at both the top

and bottom interfaces of the ferroelectric layer to suppress oxygen vacancy migration and interface defect formation. The middle panel of Fig. 3d presents a high-resolution TEM image clearly showing the orthorhombic-phase HZO film sandwiched between TiN electrodes, where the DSLs appear as a distinct interfacial layer, confirming effective defect control. The right panel of Fig. 3d shows endurance and retention characteristics, where the memory window remains above 200 mV even after 10^{11} write and 10^{13} read cycles followed by 125 °C baking, and retention extrapolation indicates > 10 years of stability at 125 °C. These results demonstrate that the defect-engineered trench-type FeRAM achieves exceptional endurance, thermal stability, and array-level reliability, meeting the stringent requirements to replace eFlash in industrial eNVM applications.

3.2 Ferroelectric tunnel junction (FTJ)

The FTJ is a two-terminal device in which a ferroelectric thin film is sandwiched between TE and BE, optionally with an additional insulating or semiconducting interlayer (IL), as shown in Fig. 4a. Current transport occurs via quantum tunneling, and the polarization direction of the ferroelectric layer modulates the interfacial charge distribution at the electrode/ferroelectric interface. This modulation changes both the height and the width of the tunneling barrier, enabling two distinct resistance states within the same structure, a low-resistance state (LRS) and a high-resistance state (HRS). As shown in the right panel of Fig. 4a, when the polarization points toward the TE, the barrier height on the BE side increases relative to that on the TE side, and the effective tunneling barrier becomes wider, which lowers the tunneling probability and results in the HRS. Conversely, when the polarization points toward the BE, the barrier height on the BE side decreases relative to that on the TE side, and the effective tunneling barrier becomes narrower compared with the HRS, leading to the LRS. This binary distinction in resistance can be exploited for non-volatile memory applications. Because FTJs utilize polarization-controlled tunneling current, they are inherently non-volatile, offer excellent scalability to the ultrathin regime, and operate at low energy. Also, their two-terminal configuration is highly compatible with crossbar array architectures, making FTJs promising candidates for high-density non-volatile memories and in-memory computing devices. At the microscopic level, the polarization of the ferroelectric layer induces bound charges at the interfaces, which interact with the finite screening length of the electrodes, asymmetry in metal work functions, and interfacial dipoles to modulate the electrostatic potential barrier. Since the tunneling probability depends exponentially on both barrier height and width, polarization reversal alone can induce a substantial change in the tunneling current.

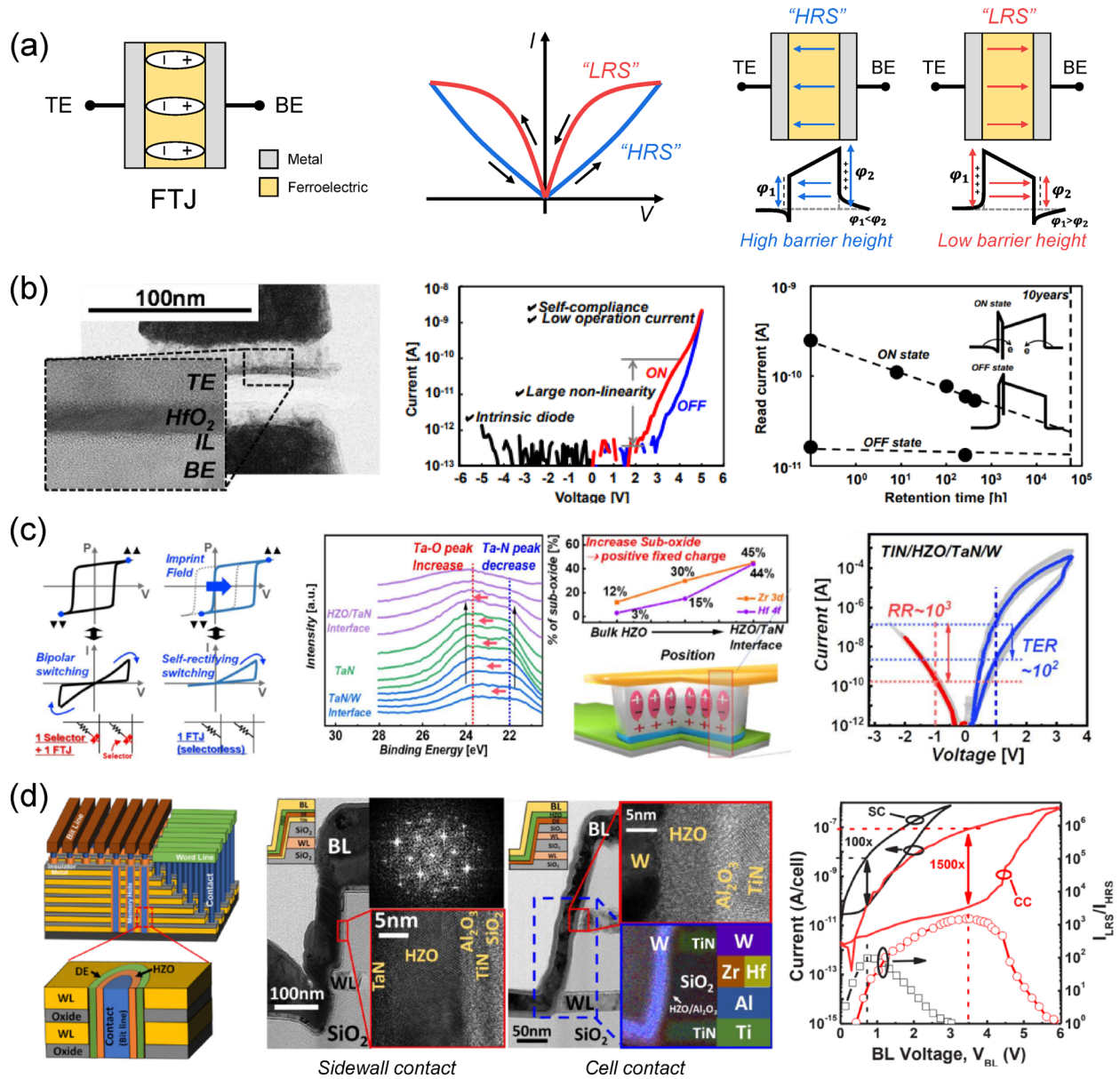


Fig. 4 FTJ. **a** Schematics of MFM FTJ with I - V characteristics for high/low resistance states and corresponding resistance changes due to barrier height variation. **b** Cross-sectional TEM image of an HfO_2 -based resistive switch showing a FTJ with intrinsic diode property (left panel), schematic illustration of current-voltage characteristics demonstrating low operation current and self-rectification (middle panel), and retention characteristics over time for different states (right panel).

Reproduced with permission from Ref [73]. Copyright 2016, IEEE. **c** Comparison of switching characteristics between conventional bipolar switching and self-rectifying switching in HZO-based FTJs (left panel), highlighting the imprint field effect. X-ray photoelectron spectroscopy (XPS) depth profile for Ta 4f, the relative sub-oxide portions of HfO_2 and ZrO_2 obtained from the Hf 4f and Zr 3d XPS spectra across the bulk HZO to the HZO/TaN interface, and a schematic of the TiN/HZO/TaN/W stack (middle panel), implying that V_o^{++} , which induces the imprint field, is generated as the TaN layer reacts with the HZO layer and contributes to the built-in potential responsible for self-rectification. The electrical performance of the TiN/HZO/TaN/W FTJ (right panel) exhibits a high rectification ratio of $\sim 10^3$ and a large TER of $\sim 10^2$. Reproduced with permission from Ref [84]. Copyright 2021, IEEE. **d** Schematic illustration of the vertical FTJ (V-FTJ) architecture, where the ferroelectric HZO layer is conformally deposited along the sidewall of a vertical hole (left panel). Cross-sectional TEM images of the sidewall contact (SC) and cell contact (CC) structures are shown, respectively, while the FFT image demonstrates the crystallinity of the HZO and the EDX analysis verifies the fabricated structure and elemental composition (middle panel). Electrical measurements of the I - V curves and $I_{\text{LRS}}/I_{\text{HRS}}$ ratios for SC and CC structures, demonstrating a high on/off ratio of 1500x for CC and a self-rectifying ratio of 100x for SC, thereby enabling selectorless operation in 3D architectures (right panel). Reproduced with permission from Ref [85]. Copyright 2023, IEEE

Although polarization-controlled tunneling was first reported by Esaki et al. in 1971 [70], the first solid-state FTJs were not realized until 2012 [71], mainly due to the difficulty of fabricating high-quality ferroelectric films only a few nanometers thick with pristine interfaces. This challenge was addressed by advances in ALD and the discovery of ferroelectricity in hafnia-based films [38, 40], enabling CMOS-compatible FTJ fabrication and driving rapid progress in device research and applications. In 2016, Fujii et al. demonstrated the first hafnia-based FTJ using a TiN/Si:HfO₂ (~5 nm)/TiN stack [72]. The left panel of Fig. 4b presents a cross-sectional TEM image revealing the device structure, where the ferroelectric HfO₂ layer and the IL are clearly identified between the TE and BE. The middle panel of Fig. 4b shows the current–voltage (*I*–*V*) characteristics, indicating a distinct conductance change upon polarization reversal due to modulation of the tunneling barrier. Notably, the device exhibited a self-rectifying characteristic without an external selector, which is a key advantage for direct crossbar integration. The right panel of Fig. 4b shows that the on-current gradually decreases over time, possibly due to electron trapping during retention. Further improvement of on-state stability could be achieved by developing FE-HfO₂ films with fewer defects, thereby suppressing electron trapping.

Following this, FTJ designs have evolved into four primary structures: MFM, metal–ferroelectric–insulator–metal (MFIM), metal–ferroelectric–semiconductor (MFS), and metal–ferroelectric–insulator–semiconductor (MFIS). In MFM and MFS devices, polarization reversal modifies the average barrier height, producing the tunneling electroresistance (TER) effect. The TER magnitude depends on the remanent polarization and the difference in electrode screening lengths [73–77]. Ambriz-Vargas et al. modeled the *I*–*V* characteristics of an MFM FTJ using a direct tunneling approach, extracting potential barriers of 1.86/2.36 eV (ON state) and 2.75/2.22 eV (OFF state) [73]. While MFM structures generally show modest TER due to small screening-length asymmetry, MFS designs can significantly enhance TER by altering semiconductor carrier concentration via polarization control, enabling accumulation, inversion, or depletion. For example, Jiao et al. achieved a TER of ~800 using an MFS FTJ with a Nb-doped SrTiO₃ electrode and a 5.8 nm-thick polycrystalline HZO layer [74]. MFIM and MFIS structures insert an insulating layer to decouple the ferroelectric film from direct electrode contact, which reduces interface defects, improves endurance, and increases TER via barrier asymmetry [78–82]. However, the IL also introduces additional depolarization fields that can degrade retention, making IL thickness optimization essential for balancing endurance, TER, and retention.

Recent work has also utilized asymmetric internal fields for self-rectifying FTJs. The left panel of Fig. 4c schematically shows *P*–*V* loops and *I*–*V* characteristics, illustrating how inserting a TaN diffusion-barrier layer into a TiN/HZO/TaN/W stack generates a positive imprint field that shifts the *P*–*V* loop and enables self-rectifying switching without an external selector [83]. The middle panel of Fig. 4c presents the X-ray photoelectron spectroscopy (XPS) depth profile for Ta 4f, the relative suboxide portions of HfO₂ and ZrO₂ obtained from the Hf 4f and Zr 3d XPS spectra across the bulk HZO to the HZO/TaN interface, and a schematic of the TiN/HZO/TaN/W stack. The XPS analysis shows that, compared with the TaN/W interface, the HZO/TaN interface exhibits an increased Ta–O bonding peak and a decreased Ta–N bonding peak, implying that the TaN layer absorbs oxygen from the HZO layer and oxidizes into Ta_xO_y. In addition, the proportion of sub-oxides increases sharply toward the HZO/TaN interface, indicating that the TaN electrode reacts with the HZO layer and forms V_o⁺⁺ at the interface, which in turn leads to the imprint effect. In the right panel of Fig. 4c, the devices exhibit pronounced diode-like *I*–*V* asymmetry, with a rectifying ratio of ~10³ at a read voltage of ~1 V and a TER of ~10², enabling stable read operation in crossbar arrays projected up to *N* ≈ 4 k, significantly higher than the *N* ≈ 30 limit for a non-rectifying TiN/HZO/W control device.

In parallel, vertical FTJ integration has been studied to overcome the footprint limitations of planar cells [84, 85]. Vertical architecture enables extreme integration density but has challenges such as conformal deposition of ultrathin ferroelectrics on vertical sidewalls, interface preservation, and low-resistance electrode formation. Lee et al. demonstrated a NAND/NOR-based 3D-stacking vertical AFE FTJ array, as illustrated in the left panel of Fig. 4d [84]. In particular, they fabricated two types of 3D vertical structures, sidewall contact (SC) and cell contact (CC), and the middle panel of Fig. 4d presents cross-sectional TEM images of the two fabricated arrays. The memory holes were formed with a side length of 1 μm, and the SC structure was distinguished by the deposition of TiN on the sidewalls. Both AFE FTJ arrays were fabricated using HZO (Zr ≈ 90%) with an Al₂O₃ interlayer, deposited on the memory-hole sidewalls by ALD and subsequently crystallized by rapid thermal anneal (RTA). The FFT image in the middle panel of Fig. 4d confirms the crystallinity of the HZO formed on the memory-hole sidewalls, while energy-dispersive X-ray (EDX) analysis shows the elemental mapping of the fabricated arrays. Furthermore, first-order reversal curve (FORC) analysis revealed a built-in bias originating from fixed charges in the Al₂O₃ and interfacial dipoles, which resulted in a nonzero remanent polarization at 0 V and contributed to the self-rectifying behavior. The right panel of

Fig. 4d presents the I – V characteristics and the $I_{\text{LRS}}/I_{\text{HRS}}$ ratio for both SC and CC structures (line: current (A/cell); symbol: $I_{\text{LRS}}/I_{\text{HRS}}$; black: SC; red: CC). The CC suppressed HRS leakage through localized HZO crystallization and a smaller effective contact area, achieving an $I_{\text{LRS}}/I_{\text{HRS}}$ ratio of $\sim 1500\times$ (versus $\sim 100\times$ for SC) and an on-state current of ~ 100 nA per cell at a current density of ~ 83 A cm $^{-2}$, sufficient for direct sensing by peripheral circuitry. The AFE FTJ also exhibited a rectification ratio $>1000\times$, supported multilevel conductance tuning via erase-voltage adjustment, and demonstrated endurance exceeding 10^9 program/erase cycles and retention beyond 10^4 s, with extrapolated 10-year on/off ratios above $10\times$. Selector-free read margins were projected to scale up to $\sim 10^4$ crossbar lines in the CC configuration. These results highlight that optimized vertical antiferroelectric FTJ structures can combine high on/off ratios, strong rectification, and robust reliability, making them viable candidates for high-density BEOL-integrated non-volatile memories.

3.3 Ferroelectric field-effect transistor (FeFET)

A FeFET is a three-terminal transistor-type non-volatile memory device that incorporates a ferroelectric layer as the gate dielectric as shown in Fig. 5a. The polarization state of the ferroelectric layer modulates the electric field applied to the channel, thereby shifting the threshold voltage (V_{th}) and creating a memory window. Applying a positive gate voltage reduces V_{th} , corresponding to the program state, whereas applying a negative gate voltage increases V_{th} , corresponding to the erase state. The resulting V_{th} shift is opposite to that observed in conventional flash memories. In some cases, however, the program and erase states are defined following the flash memory convention, where the high- V_{th} state is labeled as program and the low- V_{th} state as erase. While this reflects a difference in how the memory states are expressed, the relationship between the polarity of the applied write voltage and the direction of the V_{th} shift in FeFETs, which originates from the polarization of the ferroelectric layer, remains unchanged. Furthermore, compared with conventional flash memories, which perform program and erase operations via Fowler–Nordheim (FN) tunneling and therefore require high voltages and long programming times [86, 87], FeFETs enable low-power operation owing to their low operating voltage and fast switching speed.

In 2011, Böschke et al. first demonstrated a CMOS-compatible FeFET utilizing a Si:HfO $_2$ ferroelectric thin film [88]. The left panel of Fig. 5b shows a cross-sectional TEM image of the gate stack of the fabricated device. The device incorporated an 8.5 nm ferroelectric layer and annealed at 1000 °C, resulting in the formation of a 1.2 nm SiO $_2$ interlayer. The right panel of

Fig. 5b shows the I_{d} – V_{g} transfer curve of a short-channel device with dimensions of 150 nm $\times$ 80 nm, achieving a memory window of up to 1 V. In addition, non-volatile data retention for 10 years was demonstrated. Müller et al. fabricated a Si:HfO $_2$ -based FeFET using 28 nm high-k metal gate (HKMG) technology, as shown in the left panel of Fig. 5c [89]. Instead of conventional PZT ferroelectric, a hafnium-based ferroelectric material with high process scalability was employed, enabling the use of a 28 nm technology node with a height-to-length ratio of approximately 1. The right panel of Fig. 5c shows a memory window of 0.9 V, achieved using a ± 5 V program/erase voltage and a pulse width of 100 ns. Dünkkel et al. reported a FeFET fabricated using 22 nm fully-depleted silicon on insulator (FDSOI) CMOS technology, scaled to a cell area of 0.025 μm^2 , as shown in the left panel of Fig. 5d [90]. This work demonstrated that FeFET memory can follow the ongoing scaling trend of logic technology while also proving scalability down to the 12 nm technology node. The right panel of Fig. 5d shows a corresponding I_{d} – V_{g} transfer curve with a memory window of 1.5 V. Furthermore, write and read operations of a 32 Mbit memory array were demonstrated, highlighting the potential for low-cost, low-power expansion into Internet of Things (IoT) applications. Bae et al. demonstrated a ferroelectric FinFET with an HZO ferroelectric layer at the 25 nm technology node, as shown in the left panel of Fig. 5e [91]. The right panel of Fig. 5e shows the corresponding I_{d} – V_{g} transfer curve, obtained under sub-nanosecond switching pulses, exceeding the nanosecond-level operation speeds reported in previous studies. This performance demonstrates the potential of this device as a memory candidate within the memory hierarchy for CPUs or cache memory, where extremely high performance is required.

Beyond scaling-focused research for logic devices, significant efforts have also been directed toward applying ferroelectric memory technology to high-density NAND Flash, referred to as ferroelectric NAND (FeNAND), leveraging its high process compatibility [92–94]. Florent et al. demonstrated a 3D macaroni NAND architecture integrating a Si-doped HfO $_2$ ferroelectric layer as the memory layer, as shown in the left panel of Fig. 5f [93]. The right panel of Fig. 5f shows the corresponding I_{d} – V_{g} transfer curve of the control gate transistor, measured by applying a 3 V pass bias to both the top and bottom selectors. The vertically stacked 3D FeFET structure exhibited a memory window of approximately 2 V, demonstrating the potential of ferroelectric memory for application in high-density NAND arrays. However, the MFS configuration of the FeFET structure inherently leads to the formation of an IL between the ferroelectric layer and the semiconductor channel during the annealing process. Because the relative permittivity of the SiO $_2$ (~ 3.9) is

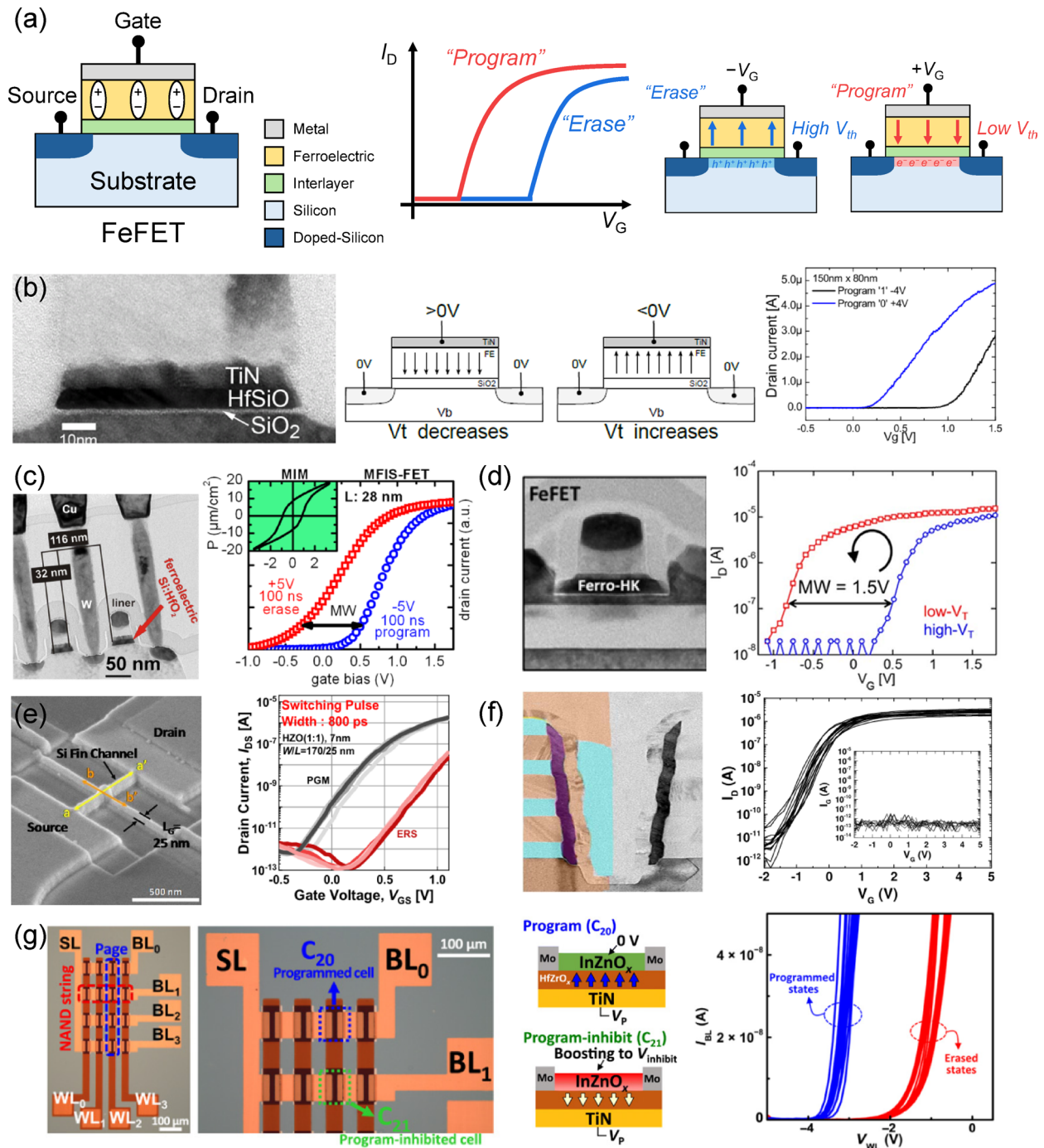


Fig. 5 FeFET. **a** Schematics of MFIS FeFET with I_D - V_G transfer curves and V_{th} shifts. **b** First reported hafnium oxide-based FeFET. Reproduced with permission from Ref [89]. Copyright 2011, IEEE. **c** 28 nm HKMG FeFET using Si:HfO₂ thin films. Reproduced with permission from Ref [90]. Copyright 2012, IEEE. **d** FeFET fabricated with 22 nm FDSOI CMOS technology. Reproduced with permission from Ref [91]. Copyright 2017, IEEE. **e** FinFET-type FeFET using HZO thin films and a 25 nm technology node. Reproduced with permission from Ref [92]. Copyright 2020, IEEE. **f** Vertical FeFET NAND with a 3D macaroni-NAND architecture. Reproduced with permission from Ref [94]. Copyright 2018, IEEE. **g** FeFET NAND with an HZO thin film and an InZnO_x oxide semiconductor channel. Reproduced with permission from Ref [100]. Copyright 2021, American Association for the Advancement of Science

significantly lower than that of the HfO₂ based ferroelectric layer (~ 20 – 40), a substantial portion of the applied gate voltage drops across the IL, thereby reducing the electric field applied to the ferroelectric layer. This field

reduction limits polarization switching and narrows the memory window, while the high electric field across the IL induces charge trapping, which further decreases the memory window and accelerates endurance degradation

due to premature dielectric breakdown. To overcome these limitations, Ni et al. replaced the conventional SiO₂ IL with high-*k* materials such as Al₂O₃, HfO₂, and La₂O₃, and analyzed the correlation between the memory window and the IL material and thickness [95]. Their results demonstrated that an optimized IL improves electric field coupling to the ferroelectric layer, thereby widening the memory window, and positively influences both retention and endurance characteristics. Xiao et al. demonstrated an HZO-based FeFET in which a SiO₂ IL was formed, followed by the deposition of a 1.5 nm ZrO₂ seed layer prior to HZO deposition [96, 97]. This approach enhanced the ferroelectric properties of the HZO film, increasing the remanent polarization and coercive voltage, improving the interface characteristics between the ferroelectric layer and the IL, and reducing electron trapping within the HZO film. As a result, a wide memory window of 2.8 V was achieved.

As another approach to addressing performance degradation caused by the IL, research has also been conducted on FeFETs employing oxide semiconductors as the channel material [98–100]. By replacing the conventional Si channel with an oxide semiconductor and employing a low-temperature process (<400 °C), the formation of the IL, which is typically produced by the oxidation of Si channel, can be suppressed. This approach also allows faster operation, reduces the operating voltage, and improves endurance. Kim et al. demonstrated a FeFET with an HZO ferroelectric layer and an indium zinc oxide (InZnO_x) channel, integrating it into a 4 × 4 FeNAND array, as shown in the left panel of Fig. 5g [99]. The middle panel of Fig. 5g shows the cross-sectional structure of the device and the polarization states during write and inhibit operations. They presented a program/erase/inhibit scheme in a junctionless transistor-based FeNAND string utilizing the oxide semiconductor channel. The right panel of Fig. 5g presents the I_d – V_g transfer curves for the program and erase states of all 16 cells in the integrated array. Furthermore, they demonstrated the scalability of this approach to 3D vertical channel structures, highlighting its potential for realizing highly integrated 3D FeNAND devices that suppress IL formation with low-temperature processing while achieving excellent device performance.

3.4 Ferroelectric memcapacitor (FeCAP)

In addition to the three ferroelectric-based memory devices discussed above, the FeCAP, a two-terminal capacitor-based device utilizing ferroelectric memory, has recently attracted attention as an in-memory computing solution. A memcapacitor is a capacitor-based device with memory functionality implemented using non-volatile memory [101–106]. The switching operation of the memory induces a capacitance difference

between memory states at the same voltage. A state exhibiting higher capacitance corresponds to the high-capacitance state (HCS), whereas a state with lower capacitance corresponds to the low-capacitance state (LCS). The ratio between the two capacitances at the read voltage (C_{HCS}/C_{LCS}) is referred to as the capacitance on/off ratio, while their difference ($C_{HCS} - C_{LCS}$) is defined as the capacitive memory window (CMW). Unlike resistive devices, a memcapacitor performs vector–matrix multiplication (VMM) operations by summing the charges induced by its capacitance. The device is read by applying a read pulse, inducing a displacement current through the charging/discharging operation according to the relation $I = C \times dV/dt$, and integrating the induced current to obtain the total charge within the applied charging/discharging voltage range. Since a larger device capacitance induces more charge for the same read voltage range, a larger capacitive memory window between the HCS and LCS allows easier distinction between the two states. Therefore, widening the capacitive memory window is a critical performance metric for this kind of device. A memcapacitor array, implemented through a crossbar architecture of two-terminal capacitor devices, can achieve high integration density. Also, as capacitor devices consume only dynamic power, they exhibit negligible static power consumption. Furthermore, their relatively high intrinsic resistance compared with metal interconnects allows them to overcome limitations of resistive devices such as IR drop and sneak-path currents without the need for a select transistor [107, 108].

FeCAPs implement non-volatile memory functionality through the polarization of the ferroelectric layer and can be classified, based on their structural configuration, into two main types: MFM and MFS structures. The structure and operating principle of an MFM-based FeCAP are illustrated in Fig. 6a. In such MFM FeCAPs, excessive oxygen vacancies at the BE interface can induce domain wall pinning during the polarization switching process, leading to asymmetric small-signal capacitance–voltage (C – V) characteristics at DC 0 V [107, 109]. Alternatively, asymmetric C – V characteristics can be implemented by employing two metals with different work functions as the TE and BE, thereby utilizing interfacial asymmetry [110, 111]. The device exploits the capacitance difference observed at DC 0 V to define the memory window between the two states, utilizing the difference in induced charge from small-signal or read-pulse within this voltage range to perform VMM operations. Since this readout method does not change the device state, the structure is identical to that of FeRAM, but it differs in that the device state is sensed by a nondestructive read operation. Hur et al. demonstrated a TiN/HZO/TiN MFM FeCAP and a 12 × 12 crossbar array [107]. The C – V characteristics of the device, shown in the left panel of Fig. 6b, reveal

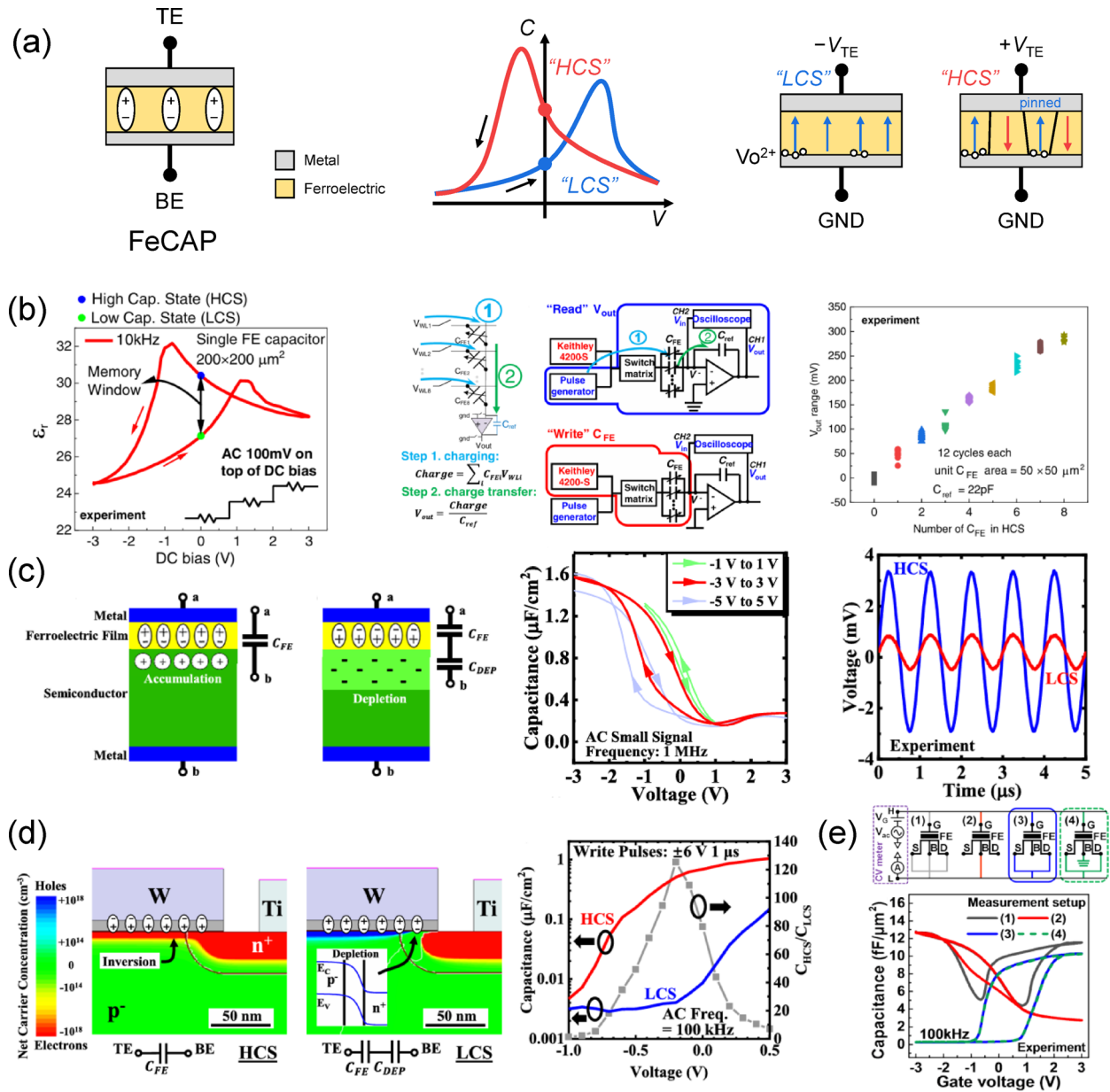


Fig. 6 FeCAP. **a** Schematics of MFM FeCAP with C–V characteristics and the mechanism of asymmetric capacitance modulation. **b** Asymmetric C–V characteristics of an MFM FeCAP (left panel), charge-based VMM operation in a FeCAP crossbar array (middle panel), and VMM output voltage as a function of the number of high-capacitance states read (right panel). Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International License from Ref [108].

Copyright 2022, John Wiley & Sons. **c** Operating principle of an accumulation-type MFS FeCAP (left panel), C–V characteristics obtained from a DC sweep (middle panel), and readout circuit output voltage for two capacitance states (right panel). Reproduced with permission from Ref [113]. Copyright 2020, IEEE. **d** Operating principle of an inversion-type MFS FeCAP (left panel) and C–V characteristics with capacitance ratio from a DC sweep (middle panel). Reproduced with permission from Ref [116]. Copyright 2023, IEEE. **e** Four capacitance measurement methods using FeFET devices (upper panel) and the corresponding C–V characteristics (bottom panel). Reproduced with permission from Ref [117]. Copyright 2023, IEEE

an on/off ratio of 1.125 at DC 0 V under 10 kHz, 100 mV AC small-signal readout, resulting from the asymmetric curve. The middle panel of Fig. 6b illustrates the charge-based VMM operation in a FeCAP crossbar array. As described earlier, charging and discharging read operations are performed through the selected WLs of the crossbar, and the charges generated in each device are

summed along the BLs. These summed charges are then converted into an output voltage through an op-amp based charge integrator circuit. The right panel of Fig. 6b presents the VMM operation results measured in the 12×12 array, showing a linear increase in the output signal with the number of HCS devices engaged in the read operation.

An MFS-based FeCAP operates in a manner similar to utilizing the ferroelectric gate stack capacitance (C_{FE}) of a FeFET, with its operating mechanism varying depending on the positions of the two electrodes. In the accumulation-type MFS FeCAP, the basic principle is derived from the C - V characteristics of a MOS capacitor, where the polarization of the ferroelectric layer shifts the flatband voltage, thereby inducing either depletion or accumulation in the underlying Si body at the same applied voltage [112, 113]. When the device is in the accumulation region, only the C_{FE} is observed, corresponding to the HCS. Conversely, when the device switches to the depletion state, the series connection of the Si depletion capacitance (C_{DEP}) and the C_{FE} results in the LCS. Zhou et al. demonstrated an accumulation-type FeCAP using an W/HfAlO_x/p-Si MFS stack [112]. The left panel of Fig. 6c illustrates the principle of capacitance modulation based on the polarization of the ferroelectric layer in an accumulation-type FeCAP. The middle panel in Fig. 6c shows the C - V characteristics over a range of DC sweep voltages, where both the typical MOS capacitor C - V profile and the flatband voltage shift induced by ferroelectric polarization, along with the resulting memory window, can be observed. Furthermore, through a DC sweep measurement from -3 to 3 V under a 1 MHz AC small signal, an on/off ratio of ~ 5 was demonstrated at 0 V. The right panel in Fig. 6c presents the transient response measured using a resistor-based readout circuit with a 100 mV, 1 MHz small-signal sine wave. The results confirm that the difference in capacitance between the two states leads to a corresponding difference in the output voltage.

The second operating mechanism of the MFS-based FeCAP, referred to as the inversion-type FeCAP, operates by configuring the gate electrode and the heavily doped region surrounding the cell as the BE [114–117]. Unlike the accumulation-type FeCAP, which employs a lightly doped Si body as the BE, this structure utilizes an ion implantation process to form a heavily doped region around the cell. This region slightly overlaps with the device area, enabling a variation in the observed capacitance depending on the switching state of the ferroelectric layer. Zhou et al. demonstrated an inversion-type FeCAP based on a W/HfAlO_x/p-Si MFS stack, where an n^+ -implanted region was formed around the p-Si body, and further experimentally validated a 32×32 crossbar array, as shown in the left panel of Fig. 6d [114, 115]. When a positive write voltage is applied to the device, the channel undergoes inversion, resulting in the HCS characterized by the C_{FE} . Conversely, when a negative voltage is applied, the channel enters depletion, and the series connection between the depletion capacitance of the overlapped depletion region C_{DEP} and C_{FE} produces the LCS. The C_{LCS} in this case is significantly smaller than that of the accumulation-type FeCAP, which utilizes the

C_{DEP} of the entire cell area, achieving a large C_{HCS}/C_{LCS} on/off ratio. The right panel of Fig. 6d shows the DC C - V curves of each state, written with ± 6 V, 1 μ s pulses and measured with a 100 kHz AC small signal, along with the on/off ratio as a function of the read voltage. A maximum on/off ratio of ~ 125 is observed near 0 V. Furthermore, this inversion-type FeCAP provides carriers for both switching directions, enabling switching operation in the nanosecond range. Under a positive write pulse, minority carrier electrons are supplied to the surface from the n^+ -doped region, whereas under a negative write pulse, holes are injected via band-to-band tunneling and trap-assisted tunneling induced by strong band bending in the overlapped region. This inversion-type FeCAP can also be implemented by utilizing the FeFET gate stack capacitance in conjunction with the n^+ -heavily doped source/drain regions as electrodes [116, 117]. Kim et al. demonstrated FeCAP operation using GlobalFoundries 28 nm node n-type FeFET [116]. Figure 6e illustrates four terminal connection configurations for the FeFET, and presents the corresponding C - V measurement results. When the source/drain region was used as the BE, the device exhibited the same operating principle as the inversion-type FeCAP, where the LCS was determined solely by the capacitance of the source/drain overlap region. This configuration achieved an on/off ratio of ~ 25 at DC 0 V under a 100 kHz AC small signal measurement, demonstrating both the feasibility and the scalability of implementing FeCAP devices using FeFET technology.

4 In-Memory computing with ferroelectric devices

4.1 Neuromorphic computing

The limitations of the conventional von Neumann architecture, such as the memory processing bottleneck and high energy consumption, have motivated the exploration of alternative computing paradigms. Neuromorphic computing, which emulates the architecture and operating principles of the human brain, offers efficient, parallel, and adaptive information processing [118–122]. A key requirement for hardware-based neuromorphic computing is the development of artificial synaptic devices capable of reliably emulating the weight modulation of biological synapses [123–127]. In hardware neural networks, the weight adjustment of artificial synapses is typically achieved by modulating the conductance of memory devices. Therefore, the potentiation and depression processes, which gradually increase or decrease the device conductance to match synaptic weight updates, are critical. Ideally, these conductance changes should be linear and symmetric, with minimal variations arising from hardware non-idealities [128]. Moreover, achieving precise weight representation and fine conductance modulation requires a multi-level capability in the device, enabling higher accuracy in neural network

computations. For stable operation, artificial synaptic devices must also exhibit high endurance for repeated conductance updates and robust nonvolatile retention characteristics to maintain weights over extended periods. Therefore, extensive research efforts have explored neuromorphic computing implementations using a wide range of emerging nonvolatile devices, each offering distinct switching mechanisms, energy efficiencies, and scalability advantages [129–138].

Ferroelectric-based devices maintain spontaneous polarization for extended periods without the need for an external voltage, offering nonvolatile characteristics with lower power consumption and faster switching speed compared to conventional charge trap flash memories. These advantages make them well suited for hardware neural network implementations that require low power, large-scale integration, and high-speed computation. By tuning the amplitude or width of the write pulse, the portion of aligned polarization can be finely controlled, enabling multi-state operation and allowing precise weight modulation in artificial neural networks [139]. In particular, HfO_2 -based ferroelectric devices exhibit high compatibility with standard CMOS fabrication processes and excellent scalability, providing strong potential for the commercialization of highly integrated arrays. In this section, recent studies on hardware implementations of artificial synapses using hafnia-based ferroelectric memory devices are discussed.

First, as discussed above, FTJs are two-terminal devices in which a thin ferroelectric layer serves as the tunneling barrier between two electrodes. Their tunneling conductance can be modulated by reversing the ferroelectric polarization, enabling non-volatile, analog-like resistance states. Specifically, potentiation is achieved by pulses that switch the polarization to lower the tunneling barrier, which increases the device conductance and drives it toward the LRS. Conversely, depression is induced by pulses that switch the polarization to raise the tunneling barrier, which decreases the device conductance and drives it toward the HRS. These characteristics render FTJs highly attractive for neuromorphic computing, where gradual and multi-level conductance modulation is essential for emulating biological synaptic weights. [140–145]. Furthermore, the tunneling-based switching mechanism offers sub-nanosecond operation speeds and ultra-low read energy, reinforcing their potential for large-scale neuromorphic processors. Figure 7a illustrates a representative FTJ device structure, composed of Ti/Au as the TE, an Al_2O_3 IL, Zr-doped HfO_2 as the ferroelectric layer, and p-type Si as the BE [140]. The inclusion of an ultrathin Al_2O_3 layer improves interface quality and helps achieve stable tunneling characteristics. The potentiation and depression characteristics were demonstrated by applying sequences of voltage pulses to gradually

modulate the device conductance using three different schemes. The first scheme used constant amplitude and width, the second used constant amplitude while varying the pulse width, and the third used constant width while gradually increasing the pulse amplitude. The middle panel of Fig. 7a presents the conductance modulation obtained with the third scheme, where a pulse width of 10 μs was maintained. For potentiation, the amplitude was increased from -4 to -8.35 V, and for depression from 4.5 to 8.85 V. As a result, the device exhibits clear and symmetric potentiation and depression characteristics across multiple conductance states, which is critical for accurate weight updates in artificial neural networks. The right panel of Fig. 7a demonstrates spike-timing-dependent plasticity (STDP), where the conductance change (ΔG) is plotted against the relative timing difference (Δt) between pre-synaptic and post-synaptic voltage spikes. Positive Δt results in potentiation, while negative Δt leads to depression. Such behavior enables unsupervised learning and associative memory functions in hardware-based neuromorphic systems, arising from the time-dependent polarization switching dynamics in the ferroelectric layer, where the overlap of voltage waveforms determines the net electric field and thus the extent of domain reorientation.

However, challenges remain in achieving linear and symmetric conductance updates over many cycles, mitigating device-to-device (D2D) variation, and ensuring long-term retention in ultra-thin ferroelectric layers. To overcome these limitations, recent research has pursued several engineering approaches. Interface engineering has been extensively explored to optimize both the ferroelectric and tunneling layers, enhancing endurance, retention, and ON/OFF ratios through careful control of interfacial quality and dielectric placement within the $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ stack [146, 147]. Such strategies not only stabilize the tunneling barrier but also improve the linearity and symmetry of conductance modulation, which is critical for reliable synaptic weight updates. In parallel, ferroelectric property recovery techniques such as thermal re-wake-up have been proposed to restore degraded polarization states, enabling the recovery of high ON/OFF ratios and stable operation in scaled FTJs [148]. From a fabrication perspective, CMOS-compatible process integration has been demonstrated to realize low-power FTJs suitable for dense neuromorphic synapse arrays without sacrificing device performance [80]. Additionally, structural and stress engineering approaches have emerged as effective tools to tailor crystallographic orientation, domain distribution, and imprint fields, resulting in improved self-rectification, reduced leakage currents, and enhanced multilevel switching behavior [149, 150]. These advancements indicate that FTJ-based synaptic devices are steadily advancing toward practical,

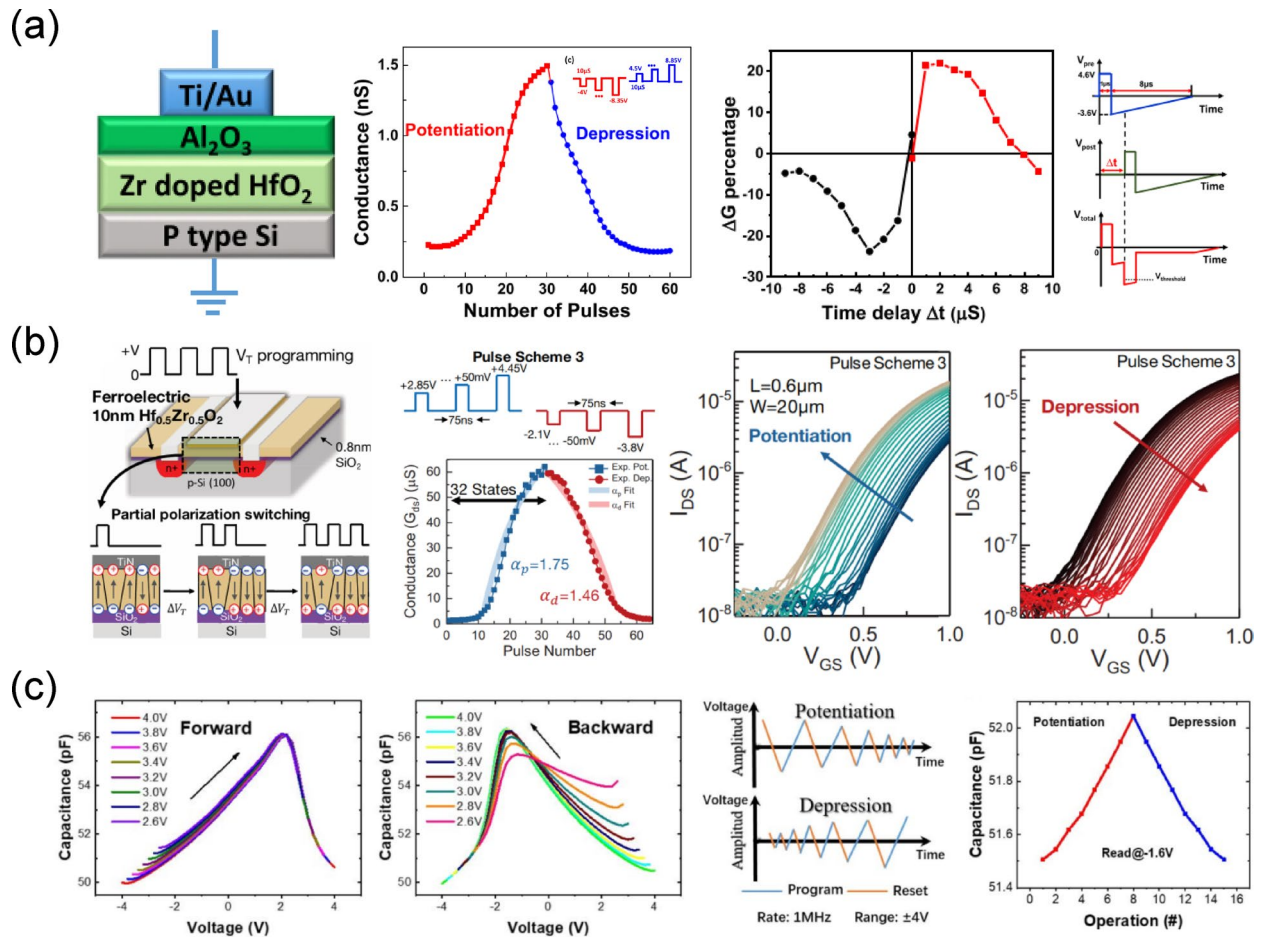


Fig. 7 **a** Schematic structure and synaptic behavior of a FTJ based on an $\text{Al}_2\text{O}_3/\text{Zr}$ -doped HfO_2 stack for neuromorphic computing. (Left panel) Cross-sectional schematic showing the Ti/Au TE Al_2O_3 IL, Zr:HfO₂ ferroelectric layer, and p-type Si substrate. (Middle panel) Potentiation and depression characteristics under repeated pulse stimulation, demonstrating conductance modulation. (Right panel) STDP response as a function of time delay (Δt) between pre- and post-synaptic spikes, along with pulse schemes for potentiation and depression. Reproduced with permission from Ref [141]. Copyright 2019, Springer Nature. **b** FeFET analog synapse for accelerating deep neural network training. (Left panel) Device schematic and mechanism of partial polarization switching in a FeFET with a 10 nm $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ ferroelectric layer and 0.8 nm SiO_2 IL. (Middle panel) Multilevel conductance modulation (32 states) obtained using a specific pulse scheme in which the gate voltage is incrementally stepped by ± 50 mV per pulse. (Right panel) Transfer characteristics (I_{DS} - V_{GS}) showing gradual changes during potentiation and depression cycles. Reproduced with permission from Ref [153]. Copyright 2017, IEEE. **c** (Left panel) C-V curves in both forward and backward sweeps under various programming voltages. (Middle panel) Schemes of the program and reset voltage waveforms for potentiation and depression, operated at a rate of 1 MHz and within a ± 4 V range. (Right panel) Capacitance modulation during repeated potentiation and depression cycles, measured at a read voltage of -1.6 V. Reproduced with permission from Ref [163]. Copyright 2019, IEEE

large-scale neuromorphic hardware, integrating high integration density, low operating energy, and analog-like weight storage in a crossbar architecture.

FeFET implements potentiation and depression by applying a write voltage to the gate, which switches the ferroelectric polarization, thereby modulating the device V_{th} and channel conductance, and consequently controlling the drain current. Compared to two-terminal structures, the three-terminal configuration offers greater immunity to sneak path currents and IR drop, enabling more accurate VMM operations. Consequently, extensive research has focused on implementing neuromorphic systems using FeFET devices [151–160]. Jerry et al.

demonstrated analog synaptic behavior using an MFIS FeFET with a 10 nm HZO layer [151]. As shown in the left panel of Fig. 7b, the channel conductance of the FeFET can be tuned by the gate voltage, and by optimizing pulse number, amplitude, interval, and width, partial polarization can be induced to achieve gradual programming. Three pulse schemes were examined, including applying identical pulses with fixed width and amplitude, applying pulses with fixed amplitude while increasing the width, and applying pulses with fixed width while increasing the amplitude. The middle panel of Fig. 7b presents the conductance modulation obtained using the third scheme. Both potentiation and depression were performed with a

constant pulse width of 75 ns, where the amplitude was increased from 2.85 to 4.45 V for potentiation and from -2.1 to -3.8 V for depression. This scheme exhibited linear modulation across 32 distinct conductance states as a function of the number of applied pulses. In addition, the non-linearity (α) of the potentiation and depression curves was analyzed as a function of the pulse number. While an ideal α is 0, the extracted non-linearity values were 1.75 for potentiation and 1.46 for depression. The right panel of Fig. 7b presents the I_d – V_g transfer curves corresponding to the potentiation and depression processes. Through careful optimization of pulse conditions, the device achieved 32 conductance states with excellent symmetry and linearity, resulting in an MNIST online learning simulation accuracy of 90%. Mulaosmanovic et al. demonstrated a synaptic device using a TiN/Si:HfO₂/SiON/Si gate stack MFIS FeFET fabricated with 28 nm HKMG technology and a 500 nm channel length [152]. Gradual channel conductance modulation was achieved through both amplitude and width modulation of the gate pulses. The single FeFET structure incorporated hafnium oxide as the active material in series with a resistor. By tuning the resistance value and the spiking duration, the device exhibited both spike transmission and synaptic plasticity through STDP. Sun et al. presented a hybrid in-situ training and inference scheme implemented with a 2 T–1FeFET weight unit cell using an HZO FeFET [153]. In this approach, during training the volatile gate voltage of the FeFET was used to represent the least significant bit, while during inference the nonvolatile polarization state stored information corresponding to the most significant bit. Their device exhibited multi-level conductance with 64 states, high symmetry, and linearity in conductance updates, achieving 97.3% recognition accuracy on the MNIST dataset. Seo et al. demonstrated a junctionless-type FeFET synaptic transistor with a FinFET structure incorporating an 8.5 nm HZO layer [154]. By applying identical pulses, the device achieved 32 distinct conductance states, enabling an MNIST recognition accuracy of 80%. Although the recognition rate was lower compared to other works due to the use of a non-optimized pulse scheme, this study highlighted the significance of applying a cost-effective, logic-process-compatible synaptic device based on a junctionless FinFET structure for neuromorphic applications.

FeCAP-based synaptic devices can utilize the high integration density of the 4F² crossbar array architecture while suppressing non-idealities such as IR drop and sneak path currents thanks to the nature of a capacitor, thereby minimizing computational errors. In addition, they consume negligible static power, offering significant advantages for large-scale synaptic operations and demonstrating strong potential for neuromorphic computing applications [107, 108]. Zheng et al. demonstrated the

TiN/Al:HfO₂/TiN (MFM) FeCAP device as a synaptic device and reported a capacitive neural network based on a FeCAP crossbar array [161]. The left panel of Fig. 7c shows the C – V characteristics for the two polarization states of the MFM FeCAP under forward and backward sweeps, as well as its multilevel capacitance behavior.

The middle panel of Fig. 7c shows the program and reset voltage waveforms used to induce potentiation and depression, distinct from the pulse schemes typically adopted in conductance-based devices. Before adjusting the capacitance state, both potentiation and depression start with a reset step that initializes the polarization to $+P_r$ by applying a backward sweep with an amplitude larger than the coercive field. Potentiation is then achieved by a subsequent set step in which the polarization is reversed from $+P_r$ to $-P_r$ through a reduced negative program voltage in the forward direction, resulting in a gradual increase in capacitance. Depression, on the other hand, is induced by applying program voltages with progressively larger amplitudes, which further enhance the polarization and thereby decrease the capacitance. After each programming sequence, capacitance modulation is measured at -1.6 V in forward sweep mode. The right panel of Fig. 7c presents the readout results for potentiation and depression performed under a ± 4 V range at 1 MHz, demonstrating symmetric and nearly linear modulation across multiple states. Building on this, the FeCAP crossbar array was integrated with an op-amp-based charge integrator circuit, enabling charge-based VMM operations within the synapse array and voltage output generation in the neuron circuit. With 3-bit weight resolution, the proposed system achieved an MNIST recognition accuracy of 78.28%, demonstrating the feasibility of implementing capacitive device-based artificial neural networks using FeCAP devices and arrays.

4.2 Hardware-based security

The rapid advancement of IoT and cloud computing technologies has led to widespread interconnection among edge devices, enabling the exchange of vast amounts of data across networks [162–164]. While such interconnectivity has driven automation and substantial efficiency gains across diverse industries, it has also exposed sensitive information, including personal data in healthcare and finance, to heightened hacking risks, thereby increasing the demand for more robust and reliable security and encryption technologies beyond conventional software-based approaches [165, 166]. Recently, research efforts have increasingly focused on exploiting the intrinsic entropy of hardware for encryption and random number generation. Representative hardware-based security primitives include the true random number generator (TRNG), which utilizes hardware entropy

to produce random numbers, cryptographic keys, and ciphers [167], and the PUF, which exploits inherent process variations from semiconductor fabrication process to generate unique device fingerprints and corresponding challenge–response pairs (CRPs) [168, 169]. As edge devices continue to scale down in size, conventional hardware security solutions that rely on large-area, complex, and high-power circuitry face intrinsic limitations. This scaling trend has increased the demand for compact, energy-efficient hardware security architectures, thereby stimulating growing interest in various NVM-based approaches. [167, 170–181]. Ferroelectric memory-based devices, with their non-volatile characteristics that enable long-term data retention without external power, low-power operation, excellent compatibility with existing CMOS processes, and proven scalability, present strong potential for next-generation hardware security systems.

Mulaosmanovic et al. demonstrated a random number generator by exploiting the cycle-to-cycle (C2C) switching variation of a poly-Si/TiN/HfO₂/SiON gate stack FeFET fabricated using a 28 nm HKMG process [182].

The left panel of Fig. 8a shows the transfer characteristics of a 30 nm gate length device in both program and erase states, exhibiting an on/off ratio exceeding 100 at 0 V. The middle panel of Fig. 8a illustrates the gate voltage pulse sequence used to extract switching variations, where the device was first set to a reference high V_{th} state by sequentially applying a 4 V program pulse and a -4 V erase pulse, followed by application of a 2.5 V program pulse. The time required for the device to switch from the high V_{th} state to the low V_{th} state was then recorded. When the channel length is scaled down to dimensions comparable to the ferroelectric domain size, the gate stack contains only one or a few switching domains. Under these conditions, a small number of dominant polarization switching events can abruptly change the device state during voltage application. To capture this behavior, 50 consecutive C2C measurements were performed, revealing that approximately 50% of switching events occurred at pulse widths near 10 μ s. The right panel of Fig. 8a presents the results obtained by applying a 2.5 V pulse with a 9.1 μ s width over 300 consecutive

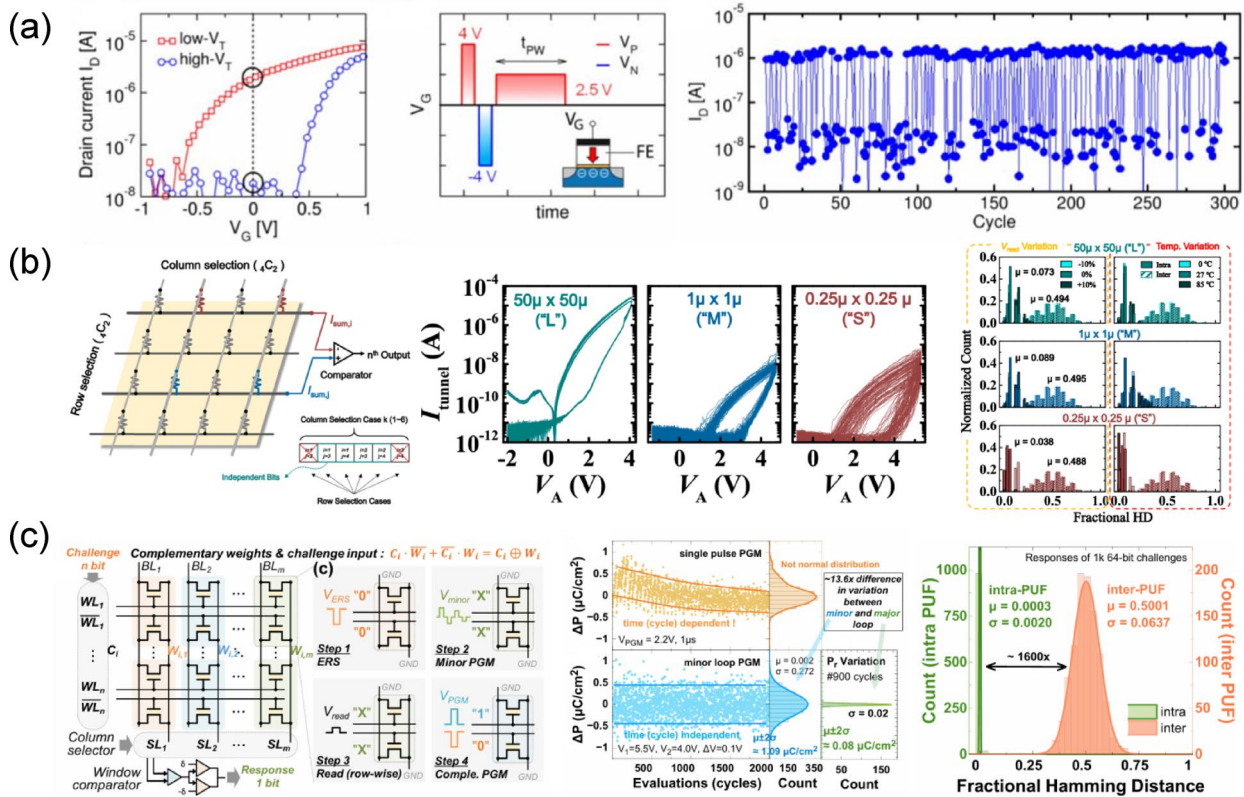


Fig. 8 a_d – V_g transfer curve of a FeFET (left panel), gate-voltage pulse scheme used to exploit C2C variation in switching pulse time (middle panel), and drain current read at $V_g = 0$ V after 300 repeated random program operations (right panel).

Reproduced with permission from Ref [184]. Copyright 2018, IEEE. **b** Schematic of a PUF design using a 4x4 FTJ crossbar array and cryptographic code generation method (left panel), I – V characteristics of FTJ devices with three different cell sizes and D2D variation across 100 devices (middle panel), and intra- and inter-HD of PUFs implemented with three different FTJ sizes with various read voltages and temperature (right panel). Reproduced with permission from Ref [185]. Copyright 2021, IEEE. **c** PUF design using a 2 T-FeFET and its enrollment and reconfiguration processes (left panel), comparison of mid-polarization state variation obtained via single-pulse and minor-loop programming (middle panel), and intra- and inter-HD of PUFs implemented using minor-loop programming in 2 T-FeFETs (right panel). Reproduced with permission from Ref [187]. Copyright 2023, IEEE

cycles, showing the device state and drain current for each cycle. Due to the 50% switching probability, the device generated random sequences of switching and non-switching events, thereby demonstrating that the inherently stochastic polarization switching of ferroelectric domains can be harnessed to generate unclonable bit sequences.

FTJ-based PUFs utilize intrinsic tunneling current variability arising from D2D differences in ferroelectric polarization states and barrier profiles to produce unique and reproducible digital fingerprints. The left panel of Fig. 8b shows a simulated 4×4 crossbar array of TiN/Hf_{0.5}Zr_{0.5}O₂/TiN FTJs, where column and row selection lines route read currents to a comparator for bit generation (Kim et al. [183]). In the middle panel of Fig. 8b, D2D variability is characterized by tunneling current measurements at a fixed read voltage for devices with lateral dimensions of 50×50 , 1×1 , and $0.25 \times 0.25 \mu\text{m}^2$. Here, smaller devices exhibit greater relative variation due to stronger local inhomogeneity effects, enhancing uniqueness. The right panel of Fig. 8b presents fractional Hamming distance (HD) distributions, showing inter-device HD values near the ideal 0.5, indicating statistical independence of responses, and intra-device HD values between 0.038 and 0.089 under $\pm 10\%$ read-voltage variation and temperature changes from 0 to 85 °C, confirming stable performance. These results highlight the potential of FTJ-based PUFs to deliver high entropy, strong repeatability, and CMOS compatibility for secure, low-power, and AI-integrated hardware systems. Yu et al. demonstrated FTJ-based PUFs using crossbar arrays with device areas of 0.6, 1.2, and $2.0 \mu\text{m}^2$ to study the effect of scaling on performance [184]. Smaller devices showed broader current distributions from stronger local defects and domain variation effects, enhancing uniqueness. Inter-HD values remained near 0.5 for all sizes, while intra-HD stayed low, confirming high reliability. This highlights that scaling can balance uniqueness and stability without added circuitry, supporting the scalability of FTJ-based PUFs for secure, low-power authentication.

Shao et al. demonstrated a PUF using a 10×10 AND-type FeFET array with a HAO/Al₂O₃ FE/IL stack, leveraging ferroelectric polarization variation induced by minor-loop program as the entropy source [185]. The PUF design and CRP generation scheme are illustrated in Fig. 8c (left panel). In the array, each column contains a cell consisting of two FeFETs, which are programmed into complementary low- V_{th} and high- V_{th} states, while complementary high and low gate voltages are applied through two WLs. In this configuration, current flows only when a low- V_{th} state is driven by a high gate voltage, effectively implementing an XOR operation. The XOR operations of individual two-FeFET cells are applied simultaneously to n complementary WLs, resulting in

2^n possible cases. The currents from individual cells are summed along with the SLs, representing both the aggregate XOR output and the total on-current. From m independent SLs, mC_2 combinations of current comparisons are performed, and a window-comparator-based scheme is used to generate the final 1-bit response. The entropy source for each cell is established by programming it into a random mid-polarization state, thereby exploiting random D2D polarization variations. This mid-polarization state is achieved through a minor-loop programming in which program and erase voltages are applied sequentially with gradually decreasing amplitudes, leading to a final random polarization configuration via domain pinning. The middle panel of Fig. 8c compares the variations obtained from 2000 repeated program operations using a single pulse versus minor-loop programming. Devices programmed via the minor-loop method exhibited approximately $13.6 \times$ greater variation. Furthermore, the high uniformity achieved through major-loop programming ensured a stable and sustainable entropy source, resulting in superior reconfigurability. The right panel of Fig. 9c presents the intra- and inter- HD distributions of the CRPs generated by the proposed PUF system, showing excellent reproducibility and uniqueness. These results highlight that the combination of high variation from ferroelectric-domain minor-loop programming and the nonlinearity introduced by XOR operations enables the realization of a highly random and robust PUF architecture. Li et al. demonstrated a reconfigurable PUF using a doped HfO₂-based 2FeFET-1C cell array fabricated in a 28 nm technology node, exploiting C2C variation as the entropy source [186]. Through charge-domain computation enabled by the capacitor in the 2FeFET-1C cell, the design achieved an energy efficiency of 1.89 fJ per bit, significantly lower than that of other state-of-the-art strong PUFs, while also delivering excellent area efficiency of $1 \mu\text{m}^2$ per cell owing to the 28 nm technology and compact cell architecture.

4.3 Associative memory

Another in-memory computing paradigm that has attracted significant attention is associative memory, also known as content-addressable memory (CAM). This architecture enables massively parallel search operations across an entire memory array in a single search operation, allowing direct data retrieval based on content rather than specific addresses [187]. Such capability makes CAM particularly suitable for data-intensive tasks, offering high-speed and energy-efficient search operations that can significantly reduce the latency and energy overhead associated with data movement in conventional computing systems. CAM architectures can be categorized into binary content-addressable memory (BCAM), which stores and searches binary data, and

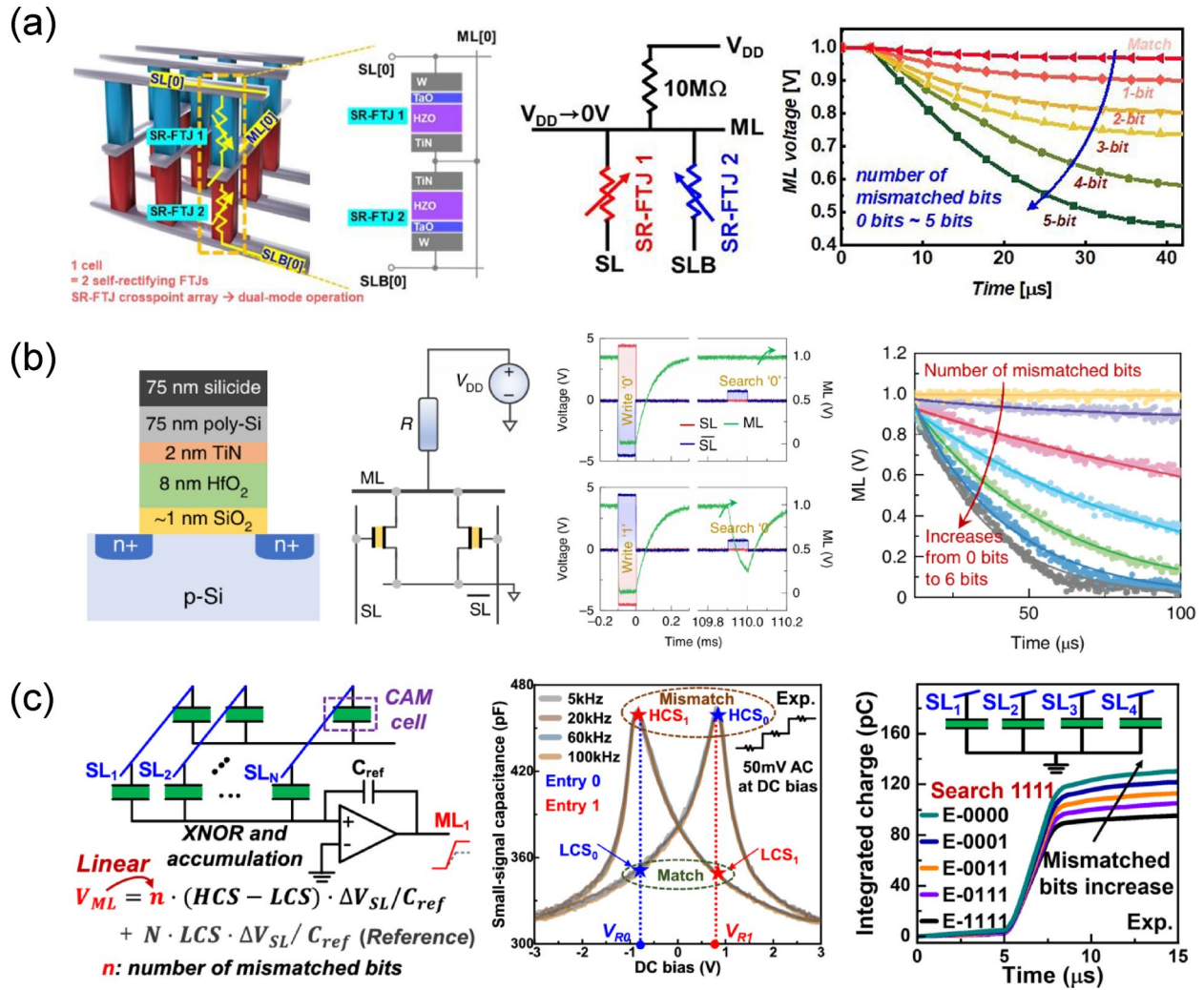


Fig. 9 a Schematic of a cross-point array composed of two self-rectifying FTJ cells (left panel), TCAM cell structure based on two self-rectifying FTJs (middle panel), and ML voltage results as a function of the number of mismatch bits (right panel).

Reproduced with permission from Ref [208]. Copyright 2023, IEEE. **b** Cross-sectional view of a FeFET device and schematic of a 2-FeFET TCAM single cell (left panel), ML voltage results during the search operation of a single TCAM cell (middle panel), and ML voltage discharge characteristics as a function of mismatch bit count in a 6-bit TCAM array (right panel). Reproduced with permission from Ref [194]. Copyright 2019, Springer Nature. **c** CAM structure using a FeCAP crossbar array (left panel), small-signal C-V characteristics of an MFM FeCAP device along with device states and search scheme (middle panel), and integrated charge results as a function of mismatch bit count in a 1 × 4 CAM array (right panel). Reproduced with permission from Ref [218]. Copyright 2024, IEEE

ternary content-addressable memory (TCAM), which incorporates an additional “X” (don’t care) bit to provide more flexibility in search operations. Traditionally, CAM has been employed in various applications such as network routing and packet classification [188–191]. More recently, its search capability has been utilized in emerging machine learning domains, including memory-augmented neural networks (MANNs), and used as a HD computing kernels [192–194]. Conventional CAM and TCAM implementations are typically based on SRAM, where a single TCAM cell consists of 16 transistors [195, 196]. However, with the growing demand for storing and searching increasingly large datasets, conventional

SRAM-based CAM faces inherent limitations arising from its large cell area and high energy consumption. To address these challenges, recent studies have explored NVM-based TCAM architectures that offer compact cell designs, low power consumption, and high-speed search capabilities [197–205]. Among these, ferroelectric memory has gained attention due to its low-power operation, high-speed, CMOS compatibility, and suitability for large-scale integration using commercially scalable technologies. In this section, we review recent advances in ferroelectric memory-based CAM research.

FTJ-based CAM combines nonvolatility with low-power operation, supporting energy-efficient, high-speed

parallel search [83, 206, 207]. As shown in Fig. 9a, Lim et al. presented an associative memory architecture based on a self-rectifying ferroelectric tunnel junction (SR-FTJ) crosspoint array, where the $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ ferroelectric layer serves as the tunneling barrier and asymmetric interfaces induce built-in rectification to suppress sneak-path currents, eliminating the need for additional selector devices [206]. The circuit schematic depicts each cell integrating two SR-FTJs connected in opposite polarities for dual-mode operation, with a load resistor on the match-line (ML) to sense current differences, enabling direct comparison between input and stored patterns. The measured ML voltage transients reveal that the voltage decay rate is strongly dependent on the number of mismatched bits, with perfect matches exhibiting minimal voltage drop and patterns with greater mismatches showing faster decay. During associative search operations, the applied input pattern produces BL currents that sum according to Ohm's and Kirchhoff's laws, and the stored pattern yielding the smallest voltage drop is identified as the correct match. The demonstration of CAM operations using a 32×32 SR-FTJ array successfully matched binary patterns without significant leakage, confirming the suitability of SR-FTJ arrays for high-density, nonvolatile, and selector-free associative memory in AI and IoT applications.

FeFET-based TCAM has also emerged as a promising solution that combines the benefits of non-volatile storage, fast switching speed, and low power operation, enabling compact cell design and high-density integration using CMOS compatible fabrication technology. Utilizing these advantages over conventional flash devices, recent research has demonstrated highly scalable TCAM implementations using FeFETs [192, 193, 208–215]. Ni et al. demonstrated a 2FeFET TCAM using an industrial 28 nm HKMG FeFET process [192]. The left panel of Fig. 9b shows the cross-sectional view of the fabricated FeFET device with an 8 nm doped HfO_2 ferroelectric layer and the schematic of a single TCAM cell. In this design, two FeFETs are programmed into complementary V_{th} states, with one device in a high V_{th} state and the other in a low V_{th} state. The search operation is carried out by applying complementary voltages, the read voltage and 0 V, to the two gates associated with the search lines (SL and SL') and prior to the search, the ML is pre-charged. In the match case, the read voltage is applied to the high V_{th} device, which remains in cut-off and prevents discharge, thereby maintaining the precharged ML voltage. In contrast, in the mismatch case, the read voltage is applied to the low V_{th} device, which turns on and discharges the ML to ground. The right panel of Fig. 9b presents search operation results for a 1×6 TCAM array, showing that the ML discharges more rapidly as the number of mismatch bits increases. Ni et al. further

investigated the linear relationship between ML discharge rate according to the mismatch bit, exploiting this HD computation capability for implementing a MANN, which was demonstrated on the Omniglot dataset. Benefiting from the compact 2FeFET cell design and a commercial technology node, their TCAM achieved an ultra small cell area of $0.15 \mu\text{m}^2$, along with low search delay and search energy, demonstrating an area and energy efficient TCAM architecture.

Lastly, research efforts have also explored implementing parallel CAM search operations using FeCAP crossbar arrays in a charge-based approach [216–218]. Xu et al. demonstrated a 1FeCAP CAM cell utilizing a TiN/HZO/TiN MFM FeCAP, with the array configuration and operating principle schematically illustrated in the left panel of Fig. 9c [216]. The C - V characteristics of the device, shown in the middle panel of Fig. 9c, reveal that for the two binary states, the device exhibits distinct HCS and LCS regions under two specific search voltage levels, from which the stored value is read using a small-signal measurement. In this scheme, a stored bit matching the search bit results in the device being in the LCS, producing a low readout charge, whereas a mismatch sets the device to the HCS, yielding a higher charge. By applying small-signal search voltages corresponding to the search data to the TEs (SLs), the resulting charges from all devices flow individually to the BEs, where they are accumulated by an op-amp integrator for sensing. Consequently, the total accumulated charge increases linearly with the number of mismatched bits, as experimentally demonstrated in a 1×4 CAM array shown in the right panel of Fig. 9c. Leveraging this linear HD computation, Xu et al. further implemented a MANN. Owing to the $4F^2$ footprint, FeCAP-based CAM enables high-density array integration, while offering low-power and high-reliability charge-based computation. Finally, when scaled to a 45 nm technology node, the design was estimated to achieve a search energy only 0.006 times that of conventional SRAM-based CAM.

5 Challenges and mitigation strategies for ferroelectric memories

A key reliability concern in ferroelectric capacitors is the degradation of the electrode/ferroelectric interface [219]. In HZO films, this interface is susceptible to defect formation, oxygen vacancy accumulation, and undesirable chemical reactions during fabrication, leading to wake-up, fatigue, and leakage current increase [220]. As shown in the left panel of Fig. 10a, Chen et al. observed that the as-fabricated TiN/HZO interface contains a parasitic TiO_xN_y layer formed during processing, which acts as a defect-rich interlayer that accelerates wake-up and fatigue [221]. They demonstrated that applying NH_3 plasma treatment to the HZO surface prior to TE

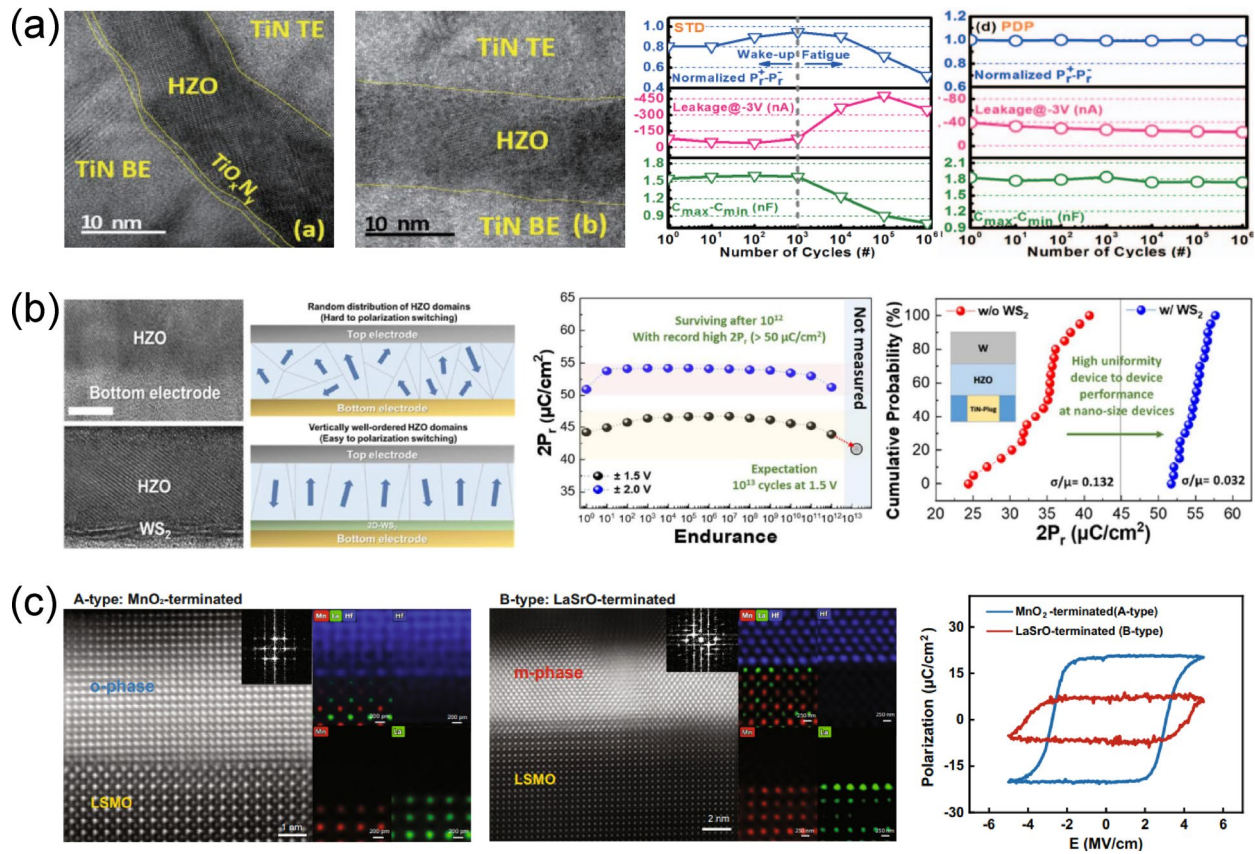


Fig. 10 **a** Cross-sectional TEM image of a TiN/HZO/TiN capacitor after NH₃ plasma treatment, showing the removal of the interfacial TiO_xN_y layer between the bottom TiN electrode and HZO, resulting in a direct TiN/HZO interface (left panel), and endurance, leakage current, and capacitance characteristics before and after NH₃ plasma treatment, demonstrating suppression of wake-up/fatigue effects, stable leakage, and negligible capacitance degradation over $> 10^{12}$ cycles (right panel).

Reproduced with permission from Ref [222]. Copyright 2017, IEEE. **b** Cross-sectional TEM images of HZO capacitors with and without a WS₂ IL (left panel), endurance characteristics over $> 10^{12}$ cycles with record-high polarization ($2P_r > 50 \mu\text{C}/\text{cm}^2$) enabled by WS₂-induced vertically ordered domains (middle panel), and cumulative probability distribution of $2P_r$ showing improved D2D uniformity at nanoscale with WS₂ insertion (right panel). Reproduced with permission from Ref [228]. Copyright 2024, IEEE. **c** Atomic-resolution STEM images of 1.5 nm-thick epitaxial HZO films on LSMO with MnO-terminated (A-type) and LaSrO-terminated (B-type) interfaces (left panel), showing orthorhombic ferroelectric and monoclinic non-ferroelectric phases, respectively, with elemental mapping (insets) confirming distinct interfacial chemistry. P - E loops (right panel) exhibit strong ferroelectric switching for the A-type and suppressed polarization for the B-type, highlighting the role of interface termination in phase stabilization. Reproduced with permission from Ref [234]. Copyright 2023, Springer Nature

deposition effectively removes or suppresses this TiO_xN_y layer, passivates oxygen vacancies, reduces defect density, and modifies interfacial bonding states. The right panel of Fig. 10a shows that this approach stabilized polarization from the initial cycles, suppressed the wake-up effect, and maintained stable switching characteristics with minimal fatigue even after 10^{10} cycles, while keeping leakage current consistently low. These results highlight that precise chemical engineering of the ferroelectric/electrode interface can substantially enhance endurance and operational reliability without the need for prolonged electrical conditioning [221–223].

Reliability in ferroelectric devices is strongly influenced by domain wall pinning and non-uniform polarization switching, which typically originate from randomly oriented ferroelectric domains in conventional

polycrystalline HZO films [96, 97, 224–226]. The left panel of Fig. 10b presents cross-sectional TEM images revealing that in conventional structures without WS₂, HZO exhibits randomly oriented domains, whereas the insertion of a monolayer WS₂ between the BE and the HZO layer leads to vertically well-aligned ferroelectric domains [227]. The middle panel of the Fig. 10b contains schematic illustrations showing how this vertical ordering facilitates easier and more uniform polarization switching by suppressing domain wall pinning. The right panel of Fig. 10b displays endurance and cumulative probability plots, indicating that WS₂-integrated HZO capacitors survive beyond 10^{12} switching cycles while maintaining a high $2P_r$ above $50 \mu\text{C}/\text{cm}^2$, along with a projected 10-year data retention at 85 °C. Moreover, the cumulative probability distribution shows a

marked improvement in uniformity, with σ/μ reduced to 0.032 compared to 0.132 without WS_2 , underscoring the scalability and reliability of this interface engineering strategy for advanced nanoscale ferroelectric memory technologies.

A major challenge for hafnia-based ferroelectrics such as HZO is the degradation of ferroelectric properties when the film thickness is reduced to the nanometer scale. Most HZO thin films are polycrystalline, containing randomly oriented grains and grain boundaries. This microstructural inhomogeneity not only introduces structural defects and phase variability but also promotes the stabilization of non-ferroelectric monoclinic or tetragonal phases, effects that become increasingly pronounced below approximately 6 nm [66]. These issues, along with enhanced depolarization fields and increased interface energy, severely degrade polarization in the ultrathin regime. Accordingly, in addition to ALD, recent work has explored high-power impulse magnetron sputtering (HiPIMS) and pulsed-laser deposition (PLD). These routes offer different trade-offs in thermal budget, impurity incorporation, and control of ultrathin phases. HiPIMS provides a highly ionized metal flux that yields dense, low-impurity HfO_2 /HZO at comparatively low substrate temperatures and allows tuning of stoichiometry and oxygen vacancies via pulse width and reactive oxygen flow, although its usable process window is tighter and wafer-level uniformity and throughput are less mature than ALD [228–232]. By contrast, PLD affords strong texture control and, on suitable single-crystal substrates, epitaxial alignment of the HZO layer with the substrate lattice, which eliminates most grain boundaries and suppresses competing phases [233–238]. As shown in Fig. 10c, Shi et al. demonstrated that the atomic termination of the underlying perovskite electrode plays a decisive role in determining the stabilized phase in ultrathin epitaxial HZO [238]. The high-resolution STEM imaging and the corresponding selected area electron diffraction (SAED) pattern reveal that growth on MnO_2 -terminated LSMO results in an Orthorhombic phase HZO film. Elemental mapping confirms the termination configuration, and the atomically ordered lattice indicates the absence of grain boundaries, enabling robust ferroelectricity even at a thickness of 1.5 nm. The middle panel of the Fig. 10c shows that when the electrode surface is LaSrO -terminated, the HZO layer instead adopts a monoclinic phase. The right panel of Fig. 10c compares the P – E hysteresis loops of the two configurations, where the orthorhombic phase film on MnO_2 termination exhibits a large remanent polarization, while the monoclinic phase film on LaSrO termination shows a markedly reduced polarization. These results highlight that precise control of interface termination during epitaxial growth can significantly

aid in stabilizing the ferroelectric phase and enhancing the performance of ultrathin HZO capacitors.

While interface-oriented approaches such as plasma treatment, 2D interlayers, and termination-controlled epitaxy have demonstrated significant benefits, they can also raise concerns including lattice mismatch at inserted interfaces and wafer-scale variability. To mitigate these limitations, other material-intrinsic strategies have also been explored to improve the reliability of ferroelectric hafnia devices. One representative route is dopant engineering, where the incorporation of aliovalent elements such as Y, Si, or Zr induces lattice distortion and charge compensation, thereby stabilizing the orthorhombic ferroelectric phase in hafnia films [239]. Another effective approach is grain size engineering, achieved by adjusting ALD cycle ratios to control the crystallite dimensions of HZO films, which intrinsically promotes ferroelectric ordering even in highly scaled devices [53, 54]. Beyond these, oxygen vacancy management through ALD process parameter tuning has also been demonstrated: modifying oxidant pulse times during ALD directly tailors the vacancy concentration profile in HfO_2 films, which in turn governs leakage and reliability characteristics [240]. Collectively, these intrinsic strategies provide scalable, CMOS-compatible pathways to enhance endurance, retention, and variability.

6 Conclusion and prospects

This review has presented the historical development of ferroelectric materials, from the discovery of the first ferroelectric compounds to the emergence of Sc-doped AlN. Among these, doped HfO_2 has emerged as the most prominent platform owing to its unique combination of scalability down to the few-nanometer regime, CMOS process compatibility, high coercive field, large bandgap, and the ability to be integrated using industry-established ALD processes with conventional electrode materials. Since the initial discovery of ferroelectricity in doped HfO_2 , extensive optimization efforts have been made to enhance its properties, including systematic engineering of dopant species and concentrations, control of HZO ALD cycle ratios, grain size engineering, annealing conditions, and electrode material and thickness tailoring. Through these advancements, the stability of the ferroelectric phase, polarization magnitude, and overall device performance have been markedly enhanced, positioning doped HfO_2 as a leading candidate for CMOS-compatible and highly scalable ferroelectric integration.

Also, we reviewed four major device classes based on hafnia ferroelectrics: FeRAM, FTJ, FeFET, and FeCAP, detailing their operating principles, initial demonstrations, and subsequent key research milestones. Building on these milestones, subsequent research has focused on scaling to advanced technology nodes, enabling

seamless integration within standard semiconductor process flows with improving endurance and retention. We also examined advanced integration strategies such as stacked vertical FTJ structures and FeNAND architectures for high-density 3D integration. The review further highlighted the growing application of hafnia-based ferroelectrics in in-memory computing, owing to their nonvolatility, fast switching speed, low operating voltage, and compatibility with advanced CMOS processes, which collectively enable energy-efficient and highly parallel computing architectures. We discussed their role in neuromorphic systems, where ferroelectric devices can emulate synaptic functionalities such as potentiation and depression, with the potential for linear and symmetric weight modulation and multi-state implementation. We also covered their application in hardware-based security, including TRNGs and PUFs, where intrinsic device variability such as domain polarization is harnessed as a robust entropy source. Additionally, we explored associative memories implemented with ferroelectric devices, achieving compact cell designs with low power consumption and high area efficiency.

Despite these advancements, hafnia-based ferroelectrics face critical reliability challenges. Wake-up, fatigue, retention loss, and interface instability remain key obstacles to long-term operational stability. Recent studies have shown that wake-up can be suppressed through optimized doping strategies in combination with capping or seed layer engineering, which minimize initial defect redistribution and stabilize the orthorhombic phase. Fatigue, arising from cumulative charge trapping and domain wall pinning under repeated cycling, can be alleviated by careful electrode selection and interface passivation layers that enhance long-term durability. Retention loss, primarily driven by depolarization fields and trap-assisted leakage, can be mitigated through dielectric scaling with ultrathin interlayers, insertion of 2D materials to improve screening, and strain engineering to reinforce temporal stability of polarization. Beyond these issue-specific solutions, interface engineering such as chemical treatments to remove defect-rich interlayers, incorporation of 2D materials for guided domain alignment, and termination-controlled epitaxial growth offers a unifying pathway to extend endurance, improve uniformity, and preserve polarization even in films thinner than 2 nm. Looking ahead, further advances in composition optimization, interface chemistry control, and microstructure engineering will be essential to fully exploit the intrinsic advantages of hafnia-based ferroelectrics. In parallel, innovations in circuit- and architecture-level design that leverage their non-volatility, fast switching, and multi-level capability could expand their application space beyond conventional memory into dense, low-power, and highly adaptive computing platforms. With the continued

convergence of materials science, device engineering, and system architecture, hafnia-based ferroelectrics are positioned to enable a new class of energy-efficient nonvolatile memories and to drive the development of secure, high-speed, and intelligent in-memory computing systems.

Abbreviations

1T-1C	One-transistor-one-capacitor
3D	Three-dimensional
A	Ampere
AFE	Antiferroelectric
Al ₂ O ₃	Aluminum oxide
ALD	Atomic layer deposition
BE	Bottom electrode
BEOL	Back-end-of-line
BL	Bit line
C	Coulomb
C2C	Cycle-to-cycle
CAM	Content-addressable memory
CC	Cell contact
C _{DEP}	Depletion capacitance
C _{FE}	Ferroelectric capacitance
C _{HCS}	Capacitance of high-capacitance state
C _{LCS}	Capacitance of low-capacitance state
CMOS	Complementary metal-oxide-semiconductor
CMW	Capacitive memory window
CRP	Challenge-response pair
CTE	Coefficient of thermal expansion
C-V	Capacitance-voltage
D2D	Device-to-device
DRAM	Dynamic random-access memory
DSL	Defect shielding layer
E	Electric field
eV	Electron volt
EDX	Energy-dispersive x-ray spectroscopy
eNVM	Embedded non-volatile memory
ERS	Erase
F	Farad
FeCAP	Ferroelectric memcapacitor
FeFET	Ferroelectric field-effect transistor
FeNAND	Ferroelectric nand
FeRAM	Ferroelectric random-access memory
FDSOI	Fully-depleted silicon on insulator
FFT	Fast fourier transform
FN	Fowler-nordheim
FORC	First-order reversal curve
FTJ	Ferroelectric tunnel junction
HAO	Hafnium aluminum oxide
HD	Hamming distance
HCS	High-capacitance state
Hf	Hafnium
HfO ₂	Hafnium oxide
HKMG	High-k metal gate
HR-TEM	High-resolution transmission electron microscopy
HRS	High-resistance state
HZO	Hafnium zirconium oxide
Hz	Hertz
I	Current
IL	Interlayer
I _{LRS}	Current of low-resistance state
I _{HRS}	Current of high-resistance state
InZnO _x	Indium zinc oxide
IoT	Internet of things
I-V	Current-voltage
J	Joule
K	Kelvin
LCS	Low-capacitance state
LRS	Low-resistance state
MANN	Memory-augmented neural network

MFIM	Metal–ferroelectric–insulator–metal
MFIS	Metal–ferroelectric–insulator–semiconductor
MFM	Metal–ferroelectric–metal
MFS	Metal–ferroelectric–semiconductor
ML	Match line
MOS	Metal–oxide–semiconductor
ms	Millisecond
nm	Nanometer
ns	Nanosecond
NVM	Non-volatile memory
P	Polarization
P–E	Polarization–electric field
PGM	Program
PL	Plate line
P _r	Remanent polarization
PUF	Physical unclonable function
P–V	Polarization–voltage
R	Resistance
RTA	Rapid thermal anneal
s	Second
SA	Sense amplifier
SC	Sidewall contact
SEM	Scanning electron microscopy
SL	Source line / search line
STDP	Spike-timing-dependent plasticity
t	Time
TaN	Tantalum nitride
Ta _x O _y	Tantalum sub-oxide
T _c	Curie temperature
TCAM	Ternary content-addressable memory
TE	Top electrode
TEM	Transmission electron microscopy
TER	Tunneling electroresistance
TiN	Titanium nitride
TRNG	True random number generator
V	Volt
V-FTJ	Vertical ferroelectric tunnel junction
VMM	Vector–matrix multiplication
V _{th}	Threshold voltage
W	Tungsten
WL	Word line
XPS	X-ray photoelectron spectroscopy
Zr	Zirconium
ZrO ₂	Zirconium oxide
ΔV	Memory window / voltage shift
μm	Micrometer
Ω	Ohm
°C	Degree celsius
%	Percent

Author contributions

H. Hwang and S. Youn equally contributed to this work. All authors read and approved of the final manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare no competing financial interests.

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References

1. A.A. Bokov, Z.-G. Ye, Recent progress in relaxor ferroelectrics with perovskite structure. *J. Mater. Sci.* **41**, 31–52 (2006). <https://doi.org/10.1007/s10853-005-5915-7>
2. F. Kubel, H. Schmid, Structure of a ferroelectric and ferroelastic monodomain crystal of the perovskite BiFeO₃. *Acta Crystallogr. Sect. B* **46**, 698–702 (1990). <https://doi.org/10.1107/S0108768190006887>
3. R.E. Cohen, Origin of ferroelectricity in perovskite oxides. *Nature* **358**, 136–138 (1992). <https://doi.org/10.1038/358136a0>
4. M.H. Park et al., Review and perspective on ferroelectric HfO₂-based thin films for memory applications. *MRS Commun.* **8**, 795–808 (2018). <https://doi.org/10.1557/mrc.2018.175>
5. J.Y. Park et al., Revival of ferroelectric memories based on emerging fluorite-structured ferroelectrics. *Adv. Mater.* **35**, 2204904 (2023). <https://doi.org/10.1002/adma.202204904>
6. A. Wang et al., Review of ferroelectric materials and devices toward ultralow voltage operation. *Adv. Funct. Mater.* **35**, 2412332 (2025). <https://doi.org/10.1002/adfm.202412332>
7. I.-J. Kim, J.-S. Lee, Ferroelectric transistors for memory and neuromorphic device applications. *Adv. Mater.* **35**, 2206864 (2023). <https://doi.org/10.1002/adma.202206864>
8. T. Mikolajick et al., From ferroelectric material optimization to neuromorphic devices. *Adv. Mater.* **35**, 2206042 (2023). <https://doi.org/10.1002/adma.202206042>
9. D. Wang et al., Ferroelectric materials for neuro-inspired computing applications. *Fundam. Res.* **4**, 1272–1291 (2024). <https://doi.org/10.1016/j.fmre.2023.04.013>
10. Y. Hu et al., Ferroelectricity based memory devices: new-generation of materials and applications. *Trans. Electr. Electron. Mater.* **24**, 271–278 (2023). <https://doi.org/10.1007/s42341-023-00445-9>
11. B. Shen et al., Review on ferroelectricity and atomic characterization of Hf_{0.5}Zr_{0.5}O₂ in FeRAM. *ACS Appl. Electron. Mater.* **7**, 4675–4702 (2025). <https://doi.org/10.1021/acsaem.5c00037>
12. J. Hwang, Y. Goh, S. Jeon, Physics, structures, and applications of fluorite-structured ferroelectric tunnel junctions. *Small* **20**, 2305271 (2024). <https://doi.org/10.1002/sml.202305271>
13. H. Mulaosmanovic et al., Ferroelectric field-effect transistors based on HfO₂: a review. *Nanotechnology* **32**, 502002 (2021). <https://doi.org/10.1088/1361-6528/ac189f>
14. A.I. Khan, A. Keshavarzi, S. Datta, The future of ferroelectric field-effect transistor technology. *Nat. Electron.* **3**, 588–597 (2020). <https://doi.org/10.1038/s41928-020-00492-7>
15. Z. Zhou et al., Ferroelectric capacitive memories: devices, arrays, and applications. *Nano Convergence* **12**, 3 (2025). <https://doi.org/10.1186/s40580-024-00463-0>
16. G. Busch, Early history of ferroelectricity. *Ferroelectrics* **74**, 267–284 (1987). <https://doi.org/10.1080/00150198708201307>
17. J. Valasek, Piezo-electric and allied phenomena in Rochelle salt. *Phys. Rev.* **17**, 475 (1921). <https://doi.org/10.1103/PhysRev.17.475>
18. J. Valasek, Piezo-electric activity of Rochelle salt under various conditions. *Phys. Rev.* **19**, 478 (1922). <https://doi.org/10.1103/PhysRev.19.478>
19. J. Valasek, Properties of Rochelle salt related to the piezo-electric effect. *Phys. Rev.* **20**, 639 (1922). <https://doi.org/10.1103/PhysRev.20.639>
20. G. Busch, P. Scherrer, Eine neue seignette-elektrische Substanz. *Naturwissenschaften* **23**, 737–737 (1935). <https://doi.org/10.1007/BF01498152>
21. A. Von Hippel et al., High dielectric constant ceramics. *Ind. Eng. Chem.* **38**, 1097–1109 (1946). <https://doi.org/10.1021/ie50443a009>
22. H.D. Megaw, Crystal structure of double oxides of the perovskite type. *Proc. Phys. Soc.* **58**, 133 (1946). <https://doi.org/10.1088/0959-5309/58/2/301>
23. W.F. Forrester, R.M. Hinde, Crystal structure of barium titanate. *Nature* **156**, 177–177 (1945). <https://doi.org/10.1038/156177b0>
24. L.D. Landau, On the theory of phase transitions. *Zh. Eksp. Teor. Fiz.* **7**, 926 (1937)

25. A.F. Devonshire, XCVI Theory of barium titanate: part I. Lond Edinburgh Dublin Philos Mag J Sci (1949). <https://doi.org/10.1080/14786444908561372>
26. A.F. Devonshire, CLX Theory of barium titanate—Part II. Lond Edinburgh Dublin Philos Mag J Sci. (1951). <https://doi.org/10.1080/14786445108561354>
27. A. Devonshire, Theory of ferroelectrics. *Adv. Phys.* **3**, 85–130 (1954). <https://doi.org/10.1080/00018735400101173>
28. G. Shirane, A. Takeda, Phase transitions in solid solutions of PbZrO_3 and PbTiO_3 (II) small concentrations of PbTiO_3 . *J. Phys. Soc. Jpn.* **7**, 5–11 (1952). <https://doi.org/10.1143/JPSJ.7.5>
29. G. Shirane, K. Suzuki, A. Takeda, Phase transitions in solid solutions of PbZrO_3 and PbTiO_3 (II) X-ray study. *J. Phys. Soc. Jpn.* **7**, 12–18 (1952). <https://doi.org/10.1143/JPSJ.7.12>
30. B. Jaffe, R.S. Roth, S. Marzullo, Piezoelectric properties of lead zirconate-lead titanate solid-solution ceramics. *J. Appl. Phys.* **25**, 809–810 (1954). <https://doi.org/10.1063/1.17217410>
31. W. Cochran, Crystal stability and the theory of ferroelectricity. *Phys. Rev. Lett.* **3**, 412 (1959). <https://doi.org/10.1080/00018736000101229>
32. G. Venkataraman, Soft modes and structural phase transitions. *Bull. Mater. Sci.* **1**, 129–170 (1979). <https://doi.org/10.1007/BF02743964>
33. R. Blinc, The soft mode concept and the history of ferroelectricity. *Ferroelectrics* **74**, 301–303 (1987). <https://doi.org/10.1080/00150198708201310>
34. D. A. Buck, Ferroelectrics for digital information storage and switching. Master's Thesis, Massachusetts Institute of Technology, Cambridge, MA 1952
35. I. M. Ross, U.S. Patent No. 2791760, 1957
36. K. Amanuma, T. Hase, Y. Miyasaka, Preparation and ferroelectric properties of $\text{SrBi}_2\text{Ta}_2\text{O}_9$ thin films. *Appl. Phys. Lett.* **66**, 221–223 (1995). <https://doi.org/10.1063/1.113140>
37. V. Garcia et al., Giant tunnel electroresistance for non-destructive readout of ferroelectric states. *Nature* **460**, 81–84 (2009). <https://doi.org/10.1038/nature08128>
38. T.S. Böscke et al., Ferroelectricity in hafnium oxide thin films. *Appl. Phys. Lett.* **99**, 102903 (2011). <https://doi.org/10.1063/1.3634052>
39. S. Fichtner, N. Wolff, F. Lofink, L. Kienle, B. Wagner, AlScN: a III-V semiconductor based ferroelectric. *J. Appl. Phys.* **125**, 114103 (2019). <https://doi.org/10.1063/1.5084945>
40. J. Muller et al., Ferroelectricity in simple binary ZrO_2 and HfO_2 . *Nano Lett.* **12**, 4318–4323 (2012). <https://doi.org/10.1021/nl302049k>
41. M. Hoffmann et al., Stabilizing the ferroelectric phase in doped hafnium oxide. *J. Appl. Phys.* **118**, 072006 (2015). <https://doi.org/10.1063/1.4927805>
42. J. Müller et al., Ferroelectric Hafnium Oxide: a CMOS-compatible and highly scalable approach to future ferroelectric memories. *IEEE Int Electron Devices Meet (IEDM)*. (2013). <https://doi.org/10.1109/IEDM.2013.6724605>
43. S. Mueller et al., Incipient ferroelectricity in Al-doped HfO_2 thin films. *Adv. Funct. Mater.* **22**, 2412–2417 (2012). <https://doi.org/10.1002/adfm.201103119>
44. J. Müller et al., Ferroelectricity in yttrium-doped hafnium oxide. *J. Appl. Phys.* **110**, 114113 (2011). <https://doi.org/10.1063/1.3667205>
45. S. Mueller et al., Ferroelectricity in Gd-doped HfO_2 thin films. *ECS J. Solid State Sci. Technol.* **1**, N123 (2012). <https://doi.org/10.1149/2.002301jss>
46. U. Schroeder et al., Impact of different dopants on the switching properties of ferroelectric hafniumoxide. *Japanese J Appl Phys* (2014). <https://doi.org/10.7567/JJAP.53.08LE02>
47. X. Tian et al., Evolution of ferroelectric HfO_2 in ultrathin region down to 3 nm. *Appl. Phys. Lett.* **112**, 102902 (2018). <https://doi.org/10.1063/1.5017094>
48. S.S. Cheema et al., Enhanced ferroelectricity in ultrathin films grown directly on silicon. *Nature* **580**, 478–482 (2020). <https://doi.org/10.1038/s41586-020-2208-x>
49. J. Junquera, P. Ghosez, Critical thickness for ferroelectricity in perovskite ultrathin films. *Nature* **422**, 506–509 (2003). <https://doi.org/10.1038/nature01501>
50. J. Müller et al., Ferroelectric $\text{Zr}_{0.5}\text{Hf}_{0.5}\text{O}_2$ thin films for nonvolatile memory applications. *Appl. Phys. Lett.* **99**, 112901 (2011). <https://doi.org/10.1063/1.3636417>
51. S. Zarubin et al., Fully ALD-grown $\text{TiN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{TiN}$ stacks: Ferroelectric and structural properties. *Appl. Phys. Lett.* **109**, 192903 (2016). <https://doi.org/10.1063/1.4966219>
52. S.J. Kim et al., Large ferroelectric polarization of $\text{TiN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{TiN}$ capacitors due to stress-induced crystallization at low thermal budget. *Appl. Phys. Lett.* **111**, 242901 (2017). <https://doi.org/10.1063/1.4995619>
53. J. Liao et al., Grain size engineering of ferroelectric Zr-doped HfO_2 for the highly scaled devices applications. *IEEE Electron Device Lett.* **40**, 1868–1871 (2019). <https://doi.org/10.1109/LED.2019.2944491>
54. Y. Peng et al., HfO_2 - ZrO_2 superlattice ferroelectric capacitor with improved endurance performance and higher fatigue recovery capability. *IEEE Electron Device Lett.* **43**, 216–219 (2021). <https://doi.org/10.1109/LED.2021.3135961>
55. J. Zhou et al., Temperature dependence of ferroelectricity in Al-doped HfO_2 featuring a high P_r of $23.7 \mu\text{C}/\text{cm}^2$. *IEEE Trans. Electron Devices* **67**, 5633–5638 (2020). <https://doi.org/10.1109/TED.2020.3032350>
56. B. Ku et al., Fast thermal quenching on the ferroelectric Al: HfO_2 thin film with record polarization density and flash memory application. in *2020 IEEE symposium on VLSI technology (VLSI)*. 2020. IEEE. <https://doi.org/10.1109/VLSITechnology18217.2020.9265024>
57. Y. Goh et al., Ultra-thin $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ thin-film-based ferroelectric tunnel junction via stress induced crystallization. *Appl. Phys. Lett.* **117**, 242901 (2020). <https://doi.org/10.1063/5.0029516>
58. Y. Lee et al., The influence of top and bottom metal electrodes on ferroelectricity of hafnia. *IEEE Trans. Electron Devices* **68**, 523–528 (2021). <https://doi.org/10.1109/TED.2020.3046173>
59. D. Bonduant, Ferroelectronic RAM memory family for critical data storage. *Ferroelectrics* **112**, 273–282 (1990). <https://doi.org/10.1080/00150199008008233>
60. T. Mikolajick, U. Schroeder, S. Slesazek, The past, the present, and the future of ferroelectric memories. *IEEE Trans. Electron Devices* **67**, 1434–1443 (2020). <https://doi.org/10.1109/TED.2020.2976148>
61. P. Polakowski et al., Ferroelectric deep trench capacitors based on Al: HfO_2 for 3D nonvolatile memory applications. In: 2014 IEEE 6th International Memory Workshop (IMW). IEEE. 2014 <https://doi.org/10.1109/IMW.2014.6849367>
62. T. Francois et al., Demonstration of BEOL-compatible ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ scaled FeRAM co-integrated with 130nm CMOS for embedded NVM applications. In: 2019 IEEE International Electron Devices Meeting (IEDM). IEEE. 2019 <https://doi.org/10.1109/IEDM19573.2019.8993485>
63. J. Okuno et al., SoC compatible 1T1C FeRAM memory array based on ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$. In: 2020 IEEE Symposium on VLSI Technology. IEEE. 2020 <https://doi.org/10.1109/VLSITechnology18217.2020.9265063>
64. J. Okuno et al., High-endurance and low-voltage operation of 1T1C FeRAM arrays for nonvolatile memory application. In: 2021 IEEE International Memory Workshop (IMW). IEEE. 2021 <https://doi.org/10.1109/IMW51353.2021.9439595>
65. M. Sung et al., Low voltage and high speed 1Xnm 1T1C FE-RAM with ultra-thin 5nm HZO. In: 2021 IEEE International Electron Devices Meeting (IEDM). IEEE. 2021 <https://doi.org/10.1109/IEDM19574.2021.9720545>
66. K. Toprasertpong et al., Low operating voltage, improved breakdown tolerance, and high endurance in $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ ferroelectric capacitors achieved by thickness scaling down to 4 nm for embedded ferroelectric memory. *ACS Appl. Mater. Interfaces* **14**, 51137–51148 (2022). <https://doi.org/10.1021/acsaami.2c15369>
67. N. Haratipour et al., Hafnia-based FeRAM: A path toward ultra-high density for next-generation high-speed embedded memory. In: 2022 International Electron Devices Meeting (IEDM). IEEE. 2022 <https://doi.org/10.1109/IEDM45625.2022.10019560>
68. T. Francois et al., High-performance operation and solder reflow compatibility in BEOL-integrated 16-kb HfO_2 : Si-based 1T–1C FeRAM arrays. *IEEE Trans. Electron Devices* **69**, 2108–2114 (2022). <https://doi.org/10.1109/TED.2021.3138360>
69. Y. Sun et al., 1T1C 3D HZO FeRAM with High Retention ($>125^\circ\text{C}$) and High Endurance ($>1\text{E}13$) for Embedded Nonvolatile Memory Application. In: 2025 Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits). IEEE. 2025 <https://doi.org/10.23919/VLSITechnologyandCirc6189.2025.11074853>
70. A.L. Esaki, R.B. Laibowitz, P.J. Stiles, Polar switch. *IBM Tech. Discl. Bull.* **13**, 114–116 (1971)
71. A. Chanthbouala et al., Solid-state memories based on ferroelectric tunnel junctions. *Nat. Nanotechnol.* **7**, 101–104 (2012). <https://doi.org/10.1038/nnano.2011.213>
72. S. Fujii et al., First demonstration and performance improvement of ferroelectric HfO_2 -based resistive switch with low operation current and intrinsic diode property. In: 2016 IEEE Symposium On VLSI Technology. IEEE. 2016 <https://doi.org/10.1109/VLSIT.2016.7573413>
73. F. Ambriz-Vargas et al., A complementary metal oxide semiconductor process-compatible ferroelectric tunnel junction. *ACS Appl. Mater. Interfaces* **9**, 13262–13268 (2017). <https://doi.org/10.1021/acsaami.6b16173>
74. P. Jiao et al., Electroresistance in metal/ferroelectric/semiconductor tunnel junctions based on a $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ barrier. *Appl. Phys. Lett.* **118**, 252901 (2021). <https://doi.org/10.1063/5.0053959>

75. Y. Goh, S. Jeon, The effect of the bottom electrode on ferroelectric tunnel junctions based on CMOS-compatible HfO_2 . *Nanotechnology* **29**, 335201 (2018). <https://doi.org/10.1088/1361-6528/aac6b3>
76. Z. Li et al., Stabilizing the ferroelectric phase in HfAlO ferroelectric tunnel junction with different bottom electrodes. *IEEE Electron Device Lett.* **44**, 947–950 (2023). <https://doi.org/10.1109/LED.2023.3269070>
77. R. Athle et al., Top electrode engineering for freedom in design and implementation of ferroelectric tunnel junctions based on $\text{Hf}_{1-x}\text{Zr}_x\text{O}_2$. *ACS Appl. Electron. Mater.* **4**, 1002–1009 (2022). <https://doi.org/10.1021/acsaem.1c01181>
78. Y.-F. Chen, L.-W. Hsu, C.-W. Hu, G.-T. Lai, Y.-H. Wu, Enhanced tunneling electro-resistance ratio for ferroelectric tunnel junctions by engineering metal work function. *IEEE Electron Device Lett.* **43**, 208–211 (2022). <https://doi.org/10.1109/LED.2021.3133577>
79. J. Hwang, Y. Goh, S. Jeon, Effect of insertion of dielectric layer on the performance of Hafnia ferroelectric devices. *IEEE Trans. Electron Devices* **68**, 841–845 (2021). <https://doi.org/10.1109/TED.2020.3043728>
80. Y.-S. Kuo et al., CMOS-compatible fabrication of low-power ferroelectric tunnel junction for neural network applications. *IEEE Trans. Electron Devices* (2021). <https://doi.org/10.1109/TED.2020.3045955>
81. Y. Cao et al., Structural engineering of $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ -based ferroelectric tunneling junction for fast-speed and low-power artificial synapses. *Adv. Electron. Mater.* (2023). <https://doi.org/10.1002/aem.202201247>
82. M.C. Sulzbach et al., Blocking of conducting channels widens window for ferroelectric resistive switching in interface-engineered structural engineering of $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ -based ferroelectric tunneling junction for fast-speed and low-power artificial synapses tunnel devices. *Adv. Funct. Mater.* (2020). <https://doi.org/10.1002/adfm.202002638>
83. Y. Goh et al., High performance and self-rectifying hafnia-based ferroelectric tunnel junction for neuromorphic computing and TCAM applications. In: 2021 IEEE International Electron Devices Meeting (IEDM). IEEE. 2021 <https://doi.org/10.1109/IEDM19574.2021.9720610>
84. J.-Y. Lee et al., 3D stackable vertical ferroelectric tunneling junction (V-F TJ) with on/off ratio 1500x, applicable cell current, self-rectifying ratio 1000x, robust endurance of 10^9 cycles, multilevel and demonstrated macro operation toward high-density BEOL NVMs. In: 2023 IEEE Symposium on VLSI Technology and Circuits (VLSI technology and circuits). IEEE. 2023 <https://doi.org/10.23919/VLSITechnologyandCirc57934.2023.10185163>
85. L. Chen et al., Ultra-low power structural engineering of $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ -based ferroelectric tunneling junction for fast-speed and low-power artificial synapses based ferroelectric tunnel junction synapses for hardware neural network applications. *Nanoscale* (2018). <https://doi.org/10.1039/C8NR04734K>
86. M.S. Song et al., Annealing effects on charge trap flash with TAHOS structure. *J. Semicond. Technol. Sci.* **24**, 323–331 (2024). <https://doi.org/10.5573/JSTS.2024.24.4.323>
87. H. Hwang et al., Wordline input bias scheme for neural network implementation in 3D-NAND flash. *Biomimetics*. **10**, 318 (2025). <https://doi.org/10.3390/biomimetics10050318>
88. T. S. Böscke et al., Ferroelectricity in hafnium oxide: CMOS compatible ferroelectric field effect transistors. In 2011 International Electron Devices Meeting. IEEE. 2011 <https://doi.org/10.1109/IEDM.2011.6131606>
89. J. Müller et al., Ferroelectricity in HfO_2 enables nonvolatile data storage in 28 nm HKMG. In: symposium on VLSI technology (VLSIT). (2012). IEEE. <https://doi.org/10.1109/VLSIT.2012.6242443>
90. S. Dinkel et al., A FeFET based super-low-power ultra-fast embedded NVM technology for 22nm FDSOI and beyond. In: 2017 IEEE International Electron Devices Meeting (IEDM). IEEE. 2017 <https://doi.org/10.1109/IEDM.2017.8268425>
91. H. Bae et al., Sub-ns polarization switching in 25nm FE FinFET toward post CPU and spatial-energetic mapping of traps for enhanced endurance. In: 2020 IEEE International Electron Devices Meeting (IEDM). IEEE. 2020 <https://doi.org/10.1109/IEDM13553.2020.9372076>
92. K. Florent et al., First demonstration of vertically stacked ferroelectric Al doped HfO_2 devices for NAND applications. In: 2017 Symposium on VLSI Technology. IEEE. 2017 <https://doi.org/10.23919/VLSIT.2017.7998162>
93. K. Florent et al., Vertical ferroelectric HfO_2 FET based on 3-D NAND architecture: Towards dense low-power memory. In: 2018 IEEE International Electron Devices Meeting (IEDM). IEEE. 2018 <https://doi.org/10.1109/IEDM.2018.8614710>
94. K. Banerjee et al., First demonstration of ferroelectric Si: HfO_2 based 3D FE-FET with trench architecture for dense nonvolatile memory application. In: 2021 IEEE International Memory Workshop (IMW). IEEE. 2021 <https://doi.org/10.1109/IMW51353.2021.9439620>
95. K. Ni et al., Critical role of interlayer in $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ ferroelectric FET nonvolatile memory performance. *IEEE Trans. Electron Devices* **65**, 2461–2469 (2018). <https://doi.org/10.1109/TED.2018.2829122>
96. W. Xiao et al., Performance improvement of $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ -based ferroelectric-field-effect transistors with ZrO_2 seed layers. *IEEE Electron Device Lett.* **40**, 714–717 (2019). <https://doi.org/10.1109/LED.2019.2903641>
97. W. Xiao et al., Memory window and endurance improvement of $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ -based FeFETs with ZrO_2 seed layers characterized by fast voltage pulse measurements. *Nanoscale Res. Lett.* **14**, 254 (2019). <https://doi.org/10.1186/s11671-019-3063-2>
98. F. Mo et al., Experimental demonstration of ferroelectric HfO_2 FET with ultrathin-body IGZO for high-density and low-power memory application. In: 2019 Symposium on VLSI Technology. IEEE. 2019 <https://doi.org/10.23919/VLSIT.2019.8776553>
99. M.-K. Kim, I.-J. Kim, J.-S. Lee, CMOS-compatible ferroelectric NAND flash memory for high-density, low-power, and high-speed three-dimensional memory. *Sci. Adv.* **7**, eabe1341 (2021). <https://doi.org/10.1126/sciadv.abe1341>
100. I.-J. Kim, M.-K. Kim, J.-S. Lee, Highly-scaled and fully-integrated 3-dimensional ferroelectric transistor array for hardware implementation of neural networks. *Nat. Commun.* **14**, 504 (2023). <https://doi.org/10.1038/s41467-023-36270-0>
101. K.-U. Demasius, A. Kirschen, S. Parkin, Energy-efficient memcapacitor devices for neuromorphic computing. *Nat. Electron.* **4**, 748–756 (2021). <https://doi.org/10.1038/s41928-021-00649-y>
102. S. Hwang et al., Capacitor-based synaptic devices for hardware spiking neural networks. *IEEE Electron Device Lett.* **43**, 549–552 (2022). <https://doi.org/10.1109/LED.2022.3149029>
103. D. Kwon, I.-Y. Chung, Capacitive neural network using charge-stored memory cells for pattern recognition applications. *IEEE Electron Device Lett.* **41**, 493–496 (2020). <https://doi.org/10.1109/LED.2020.2969695>
104. S. Ahn et al., Coupling-free readout scheme for memcapacitors with NAND flash structure. *IEEE Trans. Electron Devices* **71**, 4670–4676 (2024). <https://doi.org/10.1109/TED.2024.3418289>
105. S. Hwang et al., Memcapacitor crossbar array with charge trap NAND flash structure for neuromorphic computing. *Adv. Sci.* **10**, 2303817 (2023). <https://doi.org/10.1002/advs.202303817>
106. J. Yu et al., Vertically Stackable Memcapacitor Crossbar Array Based on NAND Flash Array Structure. In: 2024 IEEE International Electron Devices Meeting (IEDM). IEEE. 2024 <https://doi.org/10.1109/IEDM50854.2024.10873480>
107. J. Hur et al., Nonvolatile capacitive crossbar array for in-memory computing. *Adv. Intell. Syst.* **4**, 2100258 (2022). <https://doi.org/10.1002/aisy.202100258>
108. S. Yu et al., Nonvolatile capacitive synapse: device candidates for charge domain compute-in-memory. *IEEE Electron. Devices Mag.* **1**, 23–32 (2023). <https://doi.org/10.1109/MED.2023.3293060>
109. Y.-C. Luo et al., Non-volatile, small-signal capacitance in ferroelectric capacitors. *Appl. Phys. Lett.* **117**, 073501 (2020). <https://doi.org/10.1063/5.0018937>
110. S. Mukherjee et al., Pulse-based capacitive memory window with high non-destructive read endurance in fully BEOL compatible ferroelectric capacitors. In: 2023 International Electron Devices Meeting (IEDM). IEEE. 2023 <https://doi.org/10.1109/IEDM45741.2023.10413879>
111. S. Mukherjee et al., Capacitive memory window with non-destructive read in ferroelectric capacitors. *IEEE Electron Device Lett.* **44**, 1092–1095 (2023). <https://doi.org/10.1109/LED.2023.3278599>
112. Z. Zhou et al., A metal-insulator-semiconductor non-volatile programmable capacitor based on a HfAlO_x ferroelectric film. *IEEE Electron Device Lett.* **41**, 1837–1840 (2020). <https://doi.org/10.1109/LED.2020.3035276>
113. N. Liu et al., HfO_2 -based ferroelectric optoelectronic memcapacitors. *IEEE Electron Device Lett.* **44**, 524–527 (2023). <https://doi.org/10.1109/LED.2023.3235909>
114. Z. Zhou et al., Experimental Demonstration of An Inversion-Type Ferroelectric Capacitive Memory and its 1 kbit Crossbar Array Featuring High $C_{\text{HCS}}/C_{\text{LCS}}$, Fast Speed, and Long Retention. In: 2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits). IEEE. 2022 <https://doi.org/10.1109/VLSITechnologyandCirc46769.2022.9830291>
115. Z. Zhou et al., Inversion-type ferroelectric capacitive memory and its 1-Kbit crossbar array. *IEEE Trans. Electron Devices* **70**, 1641–1647 (2023). <https://doi.org/10.1109/TED.2023.3243556>
116. T.-H. Kim et al., Tunable non-volatile gate-to-source/drain capacitance of FeFET for capacitive synapse. *IEEE Electron Device Lett.* **44**, 1628–1631 (2023). <https://doi.org/10.1109/LED.2023.3311344>

117. O. Phadke et al., Reliability assessment of ferroelectric nvCAP for small-signal capacitive read-out. In: 2024 IEEE International Reliability Physics Symposium (IRPS). IEEE. 2024 <https://doi.org/10.1109/IRPS48228.2024.10529440>
118. A. Kumar et al., Multibit, lead-free Cs₂Ni₆ resistive random access memory with self-compliance for improved accuracy in binary neural network application. *Adv. Funct. Mater.* **34**, 2310780 (2024). <https://doi.org/10.1002/adfm.202310780>
119. Y. LeCun, Y. Bengio, G. Hinton, Deep learning. *Nature* **521**, 436–444 (2015). <https://doi.org/10.1038/nature14539>
120. J. Park et al., Implementation of convolutional neural networks in memristor crossbar arrays with binary activation and weight quantization. *ACS Appl. Mater. Interfaces* **16**, 1054–1065 (2024). <https://doi.org/10.1021/acsami.3c13775>
121. D. Marković et al., Physics for neuromorphic computing. *Nat. Rev. Phys.* **2**, 499–510 (2020). <https://doi.org/10.1038/s42254-020-0208-2>
122. S. Youn et al., Threshold learning algorithm for memristive neural network with binary switching behavior. *Neural Netw.* **176**, 106355 (2024). <https://doi.org/10.1016/j.neunet.2024.106355>
123. T.-H. Kim et al., Memristor crossbar array with enhanced device yield for in-memory vector-matrix multiplication. *ACS Appl. Electron. Mater.* **6**, 4099–4107 (2024). <https://doi.org/10.1021/acsaelm.4c00199>
124. J. Hasler, B. Marr, Finding a roadmap to achieve large neuromorphic hardware systems. *Front. Neurosci.* **7**, 118 (2013). <https://doi.org/10.3389/fnins.2013.00118>
125. C. Mahata et al., Optically tunable synaptic plasticity and memory emulation in Au-nanoparticle enhanced HfTiO₂/Al₂O₃-based photonic memristors. *Adv. Funct. Mater.* (2025). <https://doi.org/10.1002/adfm.202510663>
126. S.M. Lee et al., Inorganic p-type tellurium-based synaptic transistors: Complementary synaptic pairs with n-type devices for energy-efficient operation. *ACS Appl. Electron. Mater.* **6**, 5371–5378 (2024). <https://doi.org/10.1021/acsaelm.4c01027>
127. S. Youn et al., Programmable threshold logic implementations in a memristor crossbar array. *Nano Lett.* **24**, 3581–3589 (2024). <https://doi.org/10.1021/acs.nanolett.3c04073>
128. K. Kim et al., A comprehensive review of advanced trends: from artificial synapses to neuromorphic systems with consideration of non-ideal effects. *Front. Neurosci.* **18**, 1279708 (2024). <https://doi.org/10.3389/fnins.2024.1279708>
129. M. Prezioso et al., Training and operation of an integrated neuromorphic network based on metal-oxide memristors. *Nature* **521**, 61–64 (2015). <https://doi.org/10.1038/nature14441>
130. S. Kim et al., Overshoot-suppressed memristor crossbar array with high yield by AlO_x oxidation for neuromorphic system. *Adv. Mater. Technol.* **9**, 2400063 (2024). <https://doi.org/10.1002/admt.202400063>
131. M. Hu et al., Memristor-based analog computation and neural network classification with a dot product engine. *Adv. Mater.* **30**, 1705914 (2018). <https://doi.org/10.1002/adma.201705914>
132. P. Yao et al., Face classification using electronic synapses. *Nat. Commun.* **8**, 15199 (2017). <https://doi.org/10.1038/ncomms15199>
133. G. Kim, S. Youn, J. Park, H. Kim, Configurable kernel map implementation in memristor crossbar for convolution neural network. *Adv. Intell. Syst.* (2025). <https://doi.org/10.1002/aisy.202500032>
134. C. Li et al., Efficient and self-adaptive in-situ learning in multilayer memristor neural networks. *Nat. Commun.* **9**, 2385 (2018). <https://doi.org/10.1038/s41467-018-04484-2>
135. H. Kim et al., 4K-memristor analog-grade passive crossbar circuit. *Nat. Commun.* **12**, 5198 (2021). <https://doi.org/10.1038/s41467-021-25455-0>
136. S. Ambrogio et al., Equivalent-accuracy accelerated neural-network training using analogue memory. *Nature* **558**, 60–67 (2018). <https://doi.org/10.1038/s41586-018-0180-5>
137. K. Kim et al., Ising Solver using Weight Profile of Memristor Crossbar Array for Combinatorial Optimization. In: 2024 IEEE International Electron Devices Meeting (IEDM). IEEE. 2024 <https://doi.org/10.1109/IEDM50854.2024.10873396>
138. P. Yao et al., Fully hardware-implemented memristor convolutional neural network. *Nature* **577**, 641–646 (2020). <https://doi.org/10.1038/s41586-020-1942-4>
139. S. Oh et al., HfZrO_x-based ferroelectric synapse device with 32 levels of conductance states for neuromorphic applications. *IEEE Electron Device Lett.* **38**, 732–735 (2017). <https://doi.org/10.1109/LED.2017.2698083>
140. H. Ryu et al., Ferroelectric tunneling junctions based on aluminum oxide/zirconium-doped hafnium oxide for neuromorphic computing. *Sci. Rep.* **9**, 20383 (2019). <https://doi.org/10.1038/s41598-019-56816-x>
141. B. Max et al., Hafnia-based double-layer ferroelectric tunnel junctions as artificial synapses for neuromorphic computing. *ACS Appl. Electron. Mater.* **2**, 4023–4033 (2020). <https://doi.org/10.1021/acsaelm.0c00832>
142. R.-H. Koo et al., Proposition of adaptive read bias: a solution to overcome power and scaling limitations in ferroelectric-based neuromorphic system. *Adv. Sci.* **11**, 2303735 (2024). <https://doi.org/10.1002/adv.202303735>
143. S. Song et al., Highly stable artificial synapses based on ferroelectric tunnel junctions for neuromorphic computing applications. *Adv. Mater. Technol.* **7**, 2101323 (2022). <https://doi.org/10.1002/admt.202101323>
144. Y. Lu et al., Hf_{0.5}Zr_{0.5}O₂-based ferroelectric tunnel junction as an artificial synapse for speech recognition. *ACS Appl. Mater. Interfaces* **17**, 29847–29854 (2025). <https://doi.org/10.1021/acsami.5c01547>
145. R. Berdan et al., Low-power linear computation using nonlinear ferroelectric tunnel junction memristors. *Nat. Electron.* **3**, 259–266 (2020). <https://doi.org/10.1038/s41928-020-0405-0>
146. A. Sünbül et al., Optimizing ferroelectric and interface layers in HZO-based FTJs for neuromorphic applications. *IEEE Trans. Electron Devices* **69**, 808–815 (2021). <https://doi.org/10.1109/TED.2021.3131971>
147. J. Kim et al., Impact of HfO₂ dielectric layer placement in Hf_{0.5}Zr_{0.5}O₂-based ferroelectric tunnel junctions for neuromorphic applications. *Adv. Mater. Technol.* **9**, 2400050 (2024). <https://doi.org/10.1002/admt.202400050>
148. Y. Yang et al., Promote Hf_{0.5}Zr_{0.5}O₂ FTJs ON/OFF by thermal awake-up operation for neuromorphic computing. *IEEE Electron Device Lett.* (2023). <https://doi.org/10.1109/LED.2023.3336396>
149. Y. Goh et al., Selector-less ferroelectric tunnel junctions by stress engineering and an imprinting effect for high-density cross-point synapse arrays. *ACS Appl. Mater. Interfaces* **13**, 59422–59430 (2021). <https://doi.org/10.1021/acsami.1c14952>
150. Y. Cao et al., Structural engineering of H_{0.5}Z_{0.5}O₂-based ferroelectric tunneling junction for fast-speed and low-power artificial synapses. *Adv. Electron. Mater.* **9**, 2201247 (2023). <https://doi.org/10.1002/aeml.202201247>
151. M. Jerry et al., Ferroelectric FET analog synapse for acceleration of deep neural network training. In: 2017 IEEE International Electron Devices Meeting (IEDM). IEEE. 2017 <https://doi.org/10.1109/IEDM.2017.8268338>
152. H. Mulaosmanovic et al., Novel ferroelectric FET based synapse for neuromorphic systems. In: 2017 Symposium on VLSI Technology. IEEE. 2017 <https://doi.org/10.23919/VLSIT.2017.7998165>
153. X. Sun et al., Exploiting hybrid precision for training and inference: A 2T-1FeFET based analog synaptic weight cell. In: 2018 IEEE International Electron Devices Meeting (IEDM). IEEE. 2018 <https://doi.org/10.1109/IEDM.2018.8614611>
154. M. Seo et al., First demonstration of a logic-process compatible junctionless ferroelectric FinFET synapse for neuromorphic applications. *IEEE Electron Device Lett.* **39**, 1445–1448 (2018). <https://doi.org/10.1109/LED.2018.2852698>
155. M. Lederer et al., Ferroelectric field effect transistors as a synapse for neuromorphic application. *IEEE Trans. Electron Devices* **68**, 2295–2300 (2021). <https://doi.org/10.1109/TED.2021.3068716>
156. G.H. Lee et al., Ferroelectric field-effect transistors for binary neural network with 3-D NAND architecture. *IEEE Trans. Electron Devices* **69**, 6438–6445 (2022). <https://doi.org/10.1109/TED.2022.3207130>
157. M. Song et al., Ferroelectric NAND for efficient hardware bayesian neural networks. *Nat. Commun.* **16**, 6879 (2025). <https://doi.org/10.1038/s41467-025-61980-y>
158. W. Shin et al., Self-curable synaptic ferroelectric FET arrays for neuromorphic convolutional neural network. *Adv. Sci.* **10**, 2207661 (2023). <https://doi.org/10.1002/adv.202207661>
159. S. Dutta et al., Supervised learning in all FeFET-based spiking neural network: opportunities and challenges. *Front. Neurosci.* **14**, 634 (2020). <https://doi.org/10.3389/fnins.2020.00634>
160. C. Chen et al., Bio-inspired neurons based on novel leaky-FeFET with ultra-low hardware cost and advanced functionality for all-ferroelectric neural network. In: 2019 Symposium on VLSI Technology. IEEE. 2019 <https://doi.org/10.23919/VLSIT.2019.8776495>
161. Q. Zheng et al., Artificial neural network based on doped HfO₂ ferroelectric capacitors with multilevel characteristics. *IEEE Electron Device Lett.* **40**, 1309–1312 (2019). <https://doi.org/10.1109/LED.2019.2921737>
162. L. Atzori, A. Iera, G. Morabito, The internet of things: a survey. *Comput. Networks* **54**, 2787–2805 (2010). <https://doi.org/10.1016/j.comnet.2010.05.010>

163. J. Lin et al., A survey on internet of things: architecture, enabling technologies, security and privacy, and applications. *IEEE Internet Things J.* **4**, 1125–1142 (2017). <https://doi.org/10.1109/JIOT.2017.2683200>
164. Z. Zhao, K.M. Barijough, A. Gerstlauer, Deepthings: distributed adaptive deep learning inference on resource-constrained iot edge clusters. *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **37**, 2348–2359 (2018). <https://doi.org/10.1109/TCAD.2018.2858384>
165. F. Hussain et al., Machine learning in IoT security: current solutions and future challenges. *IEEE Commun. Surv. Tutor.* **22**, 1686–1721 (2020). <https://doi.org/10.1109/COMST.2020.2986444>
166. R.T. Tiburski et al., Lightweight security architecture based on embedded virtualization and trust mechanisms for IoT edge devices. *IEEE Commun. Mag.* **57**, 67–73 (2019). <https://doi.org/10.1109/MCOM.2018.1701047>
167. R. Carboni, D. Ielmini, Stochastic memory devices for security and computing. *Adv. Electron. Mater.* **5**, 1900198 (2019). <https://doi.org/10.1002/aeml.201900198>
168. C. Herder et al., Physical unclonable functions and applications: a tutorial. *Proc. IEEE* **102**, 1126–1141 (2014). <https://doi.org/10.1109/JPROC.2014.2320516>
169. Y. Gao, S.F. Al-Sarawi, D. Abbott, Physical unclonable functions. *Nat. Electron.* **3**, 81–91 (2020). <https://doi.org/10.1038/s41928-020-0372-5>
170. A. Chen, A review of emerging non-volatile memory (NVM) technologies and applications. *Solid-State Electron.* **125**, 25–38 (2016). <https://doi.org/10.1016/j.sse.2016.07.006>
171. S. Balatti et al., Physical unbiased generation of random numbers with coupled resistive switching devices. *IEEE Trans. Electron Devices* **63**, 2029–2035 (2016). <https://doi.org/10.1109/TED.2016.2537792>
172. D. Yu et al., True random number generator using stochastic noise signal of memristor with variation tolerance. *Chaos Solitons Fractals* **189**, 115708 (2024). <https://doi.org/10.1016/j.chaos.2024.115708>
173. H. Jiang et al., A novel true random number generator based on a stochastic diffusive memristor. *Nat. Commun.* **8**, 882 (2017). <https://doi.org/10.1038/s41467-017-00869-x>
174. H. Hwang et al., True random number generator using memcapacitor with charge trapping layer. *IEEE Electron Device Lett.* **45**, 1464–1467 (2024). <https://doi.org/10.1109/LED.2024.3411549>
175. J. Park, H. Kim, H. Kim, Bias-independent true random number generator circuit using memristor noise signals as entropy source. *Adv. Intell. Syst.* **7**, 2400648 (2025). <https://doi.org/10.1002/aisy.202400648>
176. H. Nili et al., Hardware-intrinsic security primitives enabled by analogue state and nonlinear conductance variations in integrated memristors. *Nat. Electron.* **1**, 197–202 (2018). <https://doi.org/10.1038/s41928-018-0039-7>
177. A. Chen, Utilizing the variability of resistive random access memory to implement reconfigurable physical unclonable functions. *IEEE Electron Device Lett.* **36**, 138–140 (2014). <https://doi.org/10.1109/LED.2014.2385870>
178. D. Yu et al., Physical unclonable function with enlarged string current variation in NAND flash array by WL selection scheme. *IEEE Trans. Electron Devices* **72**, 5400–5406 (2025). <https://doi.org/10.1109/TED.2025.3597584>
179. J. Park, H. Kim, Physical unclonable function with 3D stacked memristor crossbar array using self-differential pair. *ACS Nano* **19**, 28135–28145 (2025). <https://doi.org/10.1021/acsnano.4c18621>
180. H. Jiang et al., A provable key destruction scheme based on memristive crossbar arrays. *Nat. Electron.* **1**, 548–554 (2018). <https://doi.org/10.1038/s41928-018-0146-5>
181. M. S. Song et al., Memcapacitor crossbar array with charge trapping layer for physical unclonable function in NAND flash architecture. In: 2023 International Electron Devices Meeting (IEDM). IEEE. 2023 <https://doi.org/10.1109/IEDM45741.2023.10413842>
182. H. Mulaosmanovic et al., Random number generation based on ferroelectric switching. *IEEE Electron Device Lett.* **39**, 135–138 (2018). <https://doi.org/10.1109/LED.2017.2771818>
183. S. Kim et al., Physical unclonable functions using ferroelectric tunnel junctions. *IEEE Electron Device Lett.* **42**, 816–819 (2021). <https://doi.org/10.1109/LED.2021.3075427>
184. J. Yu et al., A novel physical unclonable function (PUF) using 16×16 pure-HfO_x ferroelectric tunnel junction array for security applications. *Nanotechnology* **32**, 485202 (2021). <https://doi.org/10.1088/1361-6528/ac1dd5>
185. H. Shao et al., A novel FeFET array-based PUF: Co-optimization of entropy source and CRP generation for enhanced robustness in IoT security. In: 2023 International Electron Devices Meeting (IEDM). IEEE. 2023 <https://doi.org/10.1109/IEDM45741.2023.10413787>
186. T. Li et al., Demonstration of high-reconfigurability and low-power strong physical unclonable function empowered by FeFET cycle-to-cycle variation and charge-domain computing. *Nat. Commun.* **16**, 189 (2025). <https://doi.org/10.1038/s41467-024-55380-x>
187. K. Pagiamtzis, A. Sheikholeslami, Content-addressable memory (CAM) circuits and architectures: a tutorial and survey. *IEEE J. Solid-State Circuits* **41**, 712–727 (2006). <https://doi.org/10.1109/JSSC.2005.864128>
188. A. J. McAuley, P. Francis, Fast routing table lookup using CAMs. In: Proc. IEEE Conf. Comput. Commun. (INFOCOM). IEEE. 1993 <https://doi.org/10.1109/INFCOM.1993.253403>
189. H.J. Chao, Next generation routers. *Proc. IEEE* **90**, 1518–1558 (2002). <https://doi.org/10.1109/JPROC.2002.802001>
190. K. Zheng et al., A TCAM-based distributed parallel IP lookup scheme and performance analysis. *IEEE/ACM Trans. Netw.* **14**, 863–875 (2006). <https://doi.org/10.1109/TNET.2006.880171>
191. M. Faezipour, M. Nourani, Wire-speed TCAM-based architectures for multi-match packet classification. *IEEE Trans. Comput.* **58**, 5–17 (2008). <https://doi.org/10.1109/TC.2008.159>
192. K. Ni et al., Ferroelectric ternary content-addressable memory for one-shot learning. *Nat. Electron.* **2**, 521–529 (2019). <https://doi.org/10.1038/s41928-019-0321-3>
193. X. S. Hu et al., In-memory computing with associative memories: A cross-layer perspective. In: 2021 IEEE International Electron Devices Meeting (IEDM). IEEE. 2021 <https://doi.org/10.1109/IEDM19574.2021.9720562>
194. R. Mao et al., Experimentally validated memristive memory augmented neural network with efficient hashing and similarity search. *Nat. Commun.* **13**, 6284 (2022). <https://doi.org/10.1038/s41467-022-33629-7>
195. O. Tyshchenko, A. Sheikholeslami, Match sensing using match-line stability in content-addressable memories (CAM). *IEEE J. Solid-State Circuits* **43**, 1972–1981 (2008). <https://doi.org/10.1109/JSSC.2008.2001932>
196. I. Hayashi et al., A 250-MHz 18-Mb full ternary CAM with low-voltage matchline sensing scheme in 65-nm CMOS. *IEEE J. Solid-State Circuits* **48**, 2671–2680 (2013). <https://doi.org/10.1109/JSSC.2013.2274888>
197. J. Li et al., 1 mb 0.41 μm² 2t–2r cell nonvolatile tcam with two-bit encoding and clocked self-referenced sensing. *IEEE J. Solid-State Circuits*. (2013). <https://doi.org/10.1109/JSSC.2013.2292055>
198. S. Matsunaga et al., Fully parallel 6T-2MTJ nonvolatile TCAM with single-transistor-based self match-line discharge control. In: 2011 Symposium on VLSI Circuits-Digest of Technical Papers. IEEE. 2011
199. C.E. Graves et al., In-memory computing with memristor content addressable memories for pattern matching. *Adv. Mater.* **32**, 2003437 (2020). <https://doi.org/10.1002/adma.202003437>
200. G. Pedretti et al., Tree-based machine learning performed in-memory with memristive analog CAM. *Nat. Commun.* **12**, 5806 (2021). <https://doi.org/10.1038/s41467-021-25873-0>
201. S. Youn et al., Memristor crossbar circuit for ternary content-addressable memory with fine-tuning operation. *Adv. Intell. Syst.* **5**, 2200325 (2023). <https://doi.org/10.1002/aisy.202200325>
202. R. Yang et al., Ternary content-addressable memory with MoS₂ transistors for massively parallel data search. *Nat. Electron.* **2**, 108–114 (2019). <https://doi.org/10.1038/s41928-019-0220-7>
203. P.-H. Tseng et al., In-memory-searching architecture based on 3D-NAND technology with ultra-high parallelism. In: 2020 IEEE International Electron Devices Meeting (IEDM). IEEE. 2020 <https://doi.org/10.1109/IEDM13553.2020.9372035>
204. P.-H. Tseng et al., A hybrid in-memory-searching and in-memory-computing architecture for NVM based AI accelerator. In: 2021 Symposium on VLSI Technology. IEEE. 2021
205. H. Hwang et al., 2C-ternary content addressable memory in memcapacitor crossbar array with NAND flash structure. *Small* **21**, 2408618 (2025). <https://doi.org/10.1002/sml.202408618>
206. S. Lim et al., Dual-mode operations of self-rectifying ferroelectric tunnel junction crosspoint array for high-density integration of IoT devices. *IEEE J. Solid-State Circuits* **58**, 1860–1870 (2023). <https://doi.org/10.1109/JSSC.2023.3265667>
207. M.-C. Nguyen et al., Experimental demonstration of combination-encoding content-addressable memory of 0.75 bits per switch utilizing Hf–Zr–O ferroelectric tunnel junctions. *Mater. Horiz.* **11**, 3307–3315 (2024). <https://doi.org/10.1039/D3MH02218H>
208. A.J. Tan et al., Experimental demonstration of a ferroelectric HfO₂-based content addressable memory cell. *IEEE Electron Device Lett.* **41**, 240–243 (2019). <https://doi.org/10.1109/LED.2019.2963300>

209. X. Yin et al., Ferroelectric FETs-based nonvolatile logic-in-memory circuits. *IEEE Trans Very Large Scale Integr (VLSI) Syst.* (2018). <https://doi.org/10.1109/TVLSI.2018.2871119>
210. C. Li et al., A scalable design of multi-bit ferroelectric content addressable memory for data-centric computing. In: 2020 IEEE International Electron Devices Meeting (IEDM). IEEE, 2020 <https://doi.org/10.1109/IEDM13553.2020.9372119>
211. X. Yin et al., FeCAM: a universal compact digital and analog content addressable memory using ferroelectric. *IEEE Trans. Electron Devices* **67**, 2785–2792 (2020). <https://doi.org/10.1109/TED.2020.2994896>
212. S. Dutta et al., Lifelong learning with monolithic 3D ferroelectric ternary content-addressable memory. In: 2021 IEEE international Electron devices meeting (IEDM). IEEE, 2021 <https://doi.org/10.1109/IEDM19574.2021.9720495>
213. X. Yin et al., Ferroelectric ternary content addressable memories for energy-efficient associative search. *IEEE Trans Comput-Aided Des Integr Circuits Syst* (2022). <https://doi.org/10.1109/TCAD.2022.3197694>
214. J. Luo et al., A novel ambipolar ferroelectric tunnel FinFET based content addressable memory with ultra-low hardware cost and high energy efficiency for machine learning. In: 2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits). IEEE, 2022 <https://doi.org/10.1109/VLSITechnologyandCirc46769.2022.9830413>
215. M. Ryu et al., One-transistor ternary content-addressable memory based on localized ferroelectric switching for massive and accurate routing. *IEEE Electron Device Lett.* **45**, 144–147 (2023). <https://doi.org/10.1109/LED.2023.345412>
216. W. Xu et al., A novel small-signal ferroelectric capacitance-based content addressable memory for area-and energy-efficient lifelong learning. *IEEE Electron Device Lett.* **45**, 24–27 (2023). <https://doi.org/10.1109/LED.2023.3333849>
217. O. Phadke et al., Ternary content addressable memory with ferroelectric non-volatile capacitor. *IEEE Electron Device Lett.* **46**, 872–875 (2025). <https://doi.org/10.1109/LED.2025.3553473>
218. Z. Zhou et al., Charge-domain content addressable memory based on ferroelectric capacitive memory for reliable and energy-efficient one-shot learning. *Nat. Commun.* **16**, 8047 (2025). <https://doi.org/10.1038/s41467-025-63190-y>
219. P. Jiang et al., Wake-up effect in HfO_2 -based ferroelectric films. *Adv. Electron. Mater.* **7**, 2000728 (2021). <https://doi.org/10.1002/aem.202000728>
220. H.J. Kim et al., A study on the wake-up effect of ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ films by pulse-switching measurement. *Nanoscale* **8**, 1383–1389 (2016). <https://doi.org/10.1039/C5NR05339K>
221. K.-Y. Chen, P.-H. Chen, Y.-H. Wu, Excellent reliability of ferroelectric HfZrO_x free from wake-up and fatigue effects by NH_3 plasma treatment. In: 2017 Symposium on VLSI Circuits. IEEE, 2017 <https://doi.org/10.23919/VLSIC.2017.8008572>
222. K.-Y. Chen et al., Impact of plasma treatment on reliability performance for HfZrO_x -based metal-ferroelectric-metal capacitors. *IEEE Electron Device Lett.* **39**, 87–90 (2017). <https://doi.org/10.1109/LED.2017.2771390>
223. P. Yuan et al., Enhanced remnant polarization ($30 \mu\text{C}/\text{cm}^2$) and retention of ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ by NH_3 plasma treatment. *IEEE Electron Device Lett.* **43**, 1045–1048 (2022). <https://doi.org/10.1109/LED.2022.3178867>
224. V. Gaddam, D. Das, S. Jeon, Insertion of HfO_2 seed/dielectric layer to the ferroelectric HZO films for heightened remanent polarization in MFM capacitors. *IEEE Trans. Electron Devices* **67**, 745–7502 (2020). <https://doi.org/10.1109/TED.2019.2961208>
225. T. Onaya et al., Improvement in ferroelectricity of $\text{Hf}_x\text{Zr}_{1-x}\text{O}_2$ thin films using ZrO_2 seed layer. *Appl. Phys. Express* **10**, 081501 (2017). <https://doi.org/10.7567/APEX.10.081501>
226. Y. Qi et al., Ferroelectricity of as-deposited HZO fabricated by plasma-enhanced atomic layer deposition at 300°C by inserting TiO_2 interlayers. *Appl. Phys. Lett.* **118**, 032906 (2021). <https://doi.org/10.1063/5.0037887>
227. S. Hwang et al., Record Endurance ($>10^{12}$ Cycles), High Polarization ($2\text{Pr} > 50 \mu\text{C}/\text{cm}^2$), and 10-Year Data Retention (85°C) in HZO Capacitors with Well-Ordered Ferroelectric Domain Structures via 2D-WS2 Interface. In: 2024 IEEE International Electron Devices Meeting (IEDM). IEEE, 2024 <https://doi.org/10.1109/IEDM50854.2024.10873482>
228. H.A. Hsain et al., Many routes to ferroelectric HfO_2 : A review of current work. *J. Vac. Sci. Technol. A* **40**, 010803 (2021). <https://doi.org/10.1116/6.0001317>
229. S.T. Jaszwski et al., Impact of HiPIMS pulse width on nucleation, crystallization, microstructure, and ferroelectric properties of HfO_2 . *J. Vac. Sci. Technol. A* **42**, 023414 (2024). <https://doi.org/10.1116/6.0003307>
230. M.-J. Zhao et al., Dielectric properties of HfO_2 films prepared by HiPIMS: Influence of oxygen supply on plasma state and film properties. *Vacuum* **224**, 112174 (2024). <https://doi.org/10.1016/j.vacuum.2024.112174>
231. U. Helmersson et al., Ionized physical vapor deposition (IPVD): a review of technology and applications. *Thin Solid Films* **513**(1–2), 1–24 (2006). <https://doi.org/10.1016/j.tsf.2006.03.033>
232. J.P.B. Silva et al., Roadmap on ferroelectric hafnia—and zirconia-based materials and devices. *APL Mater.* **11**, 089201 (2023). <https://doi.org/10.1063/5.0148068>
233. S. Estandia et al., Engineering ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ thin films by epitaxial stress. *ACS Appl. Electron. Mater.* **1**, 1449–1457 (2019). <https://doi.org/10.1021/acsaem.9b00256>
234. H.Y. Yoong et al., Epitaxial ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ thin films and their implementations in memristors for brain-inspired computing. *Adv. Funct. Mater.* **28**, 1806037 (2018). <https://doi.org/10.1002/adfm.201806037>
235. J. Lyu et al., Robust ferroelectricity in epitaxial $\text{Hf}_{1/2}\text{Zr}_{1/2}\text{O}_2$ thin films. *Appl. Phys. Lett.* **113**, 082902 (2018). <https://doi.org/10.1063/1.5041715>
236. I. Fina, F. Sanchez, Epitaxial ferroelectric HfO_2 films: growth, properties, and devices. *ACS Appl. Electron. Mater.* **3**, 1530–1549 (2021). <https://doi.org/10.1021/acsaem.1c00110>
237. J. Cao et al., An overview of ferroelectric hafnia and epitaxial growth. *Phys. Status Solidi-Rapid Res. Lett.* **15**, 2100025 (2021). <https://doi.org/10.1002/pssr.202100025>
238. S. Shi et al., Interface-engineered ferroelectricity of epitaxial $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ thin films. *Nat. Commun.* **14**, 1780 (2023). <https://doi.org/10.1038/s41467-023-37560-3>
239. R. Batra et al., Dopants promoting ferroelectricity in hafnia: insights from a comprehensive chemical space exploration. *Chem. Mater.* **29**, 9102–9109 (2017). <https://doi.org/10.1021/acs.chemmater.7b02835>
240. A.S. Sokolov et al., Influence of oxygen vacancies in ALD HfO_{2-x} thin films on non-volatile resistive switching phenomena with a $\text{Ti}/\text{HfO}_{2-x}/\text{Pt}$ structure. *Appl. Surf. Sci.* **434**, 822–830 (2018). <https://doi.org/10.1016/j.apsusc.2017.11.016>

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